

- d. Spongy bone tissue
- e. Medullary or nutrient artery
- f. Proximal epiphysis
- g. Epiphyseal plate
- h. Articular cartilage
- i. Compact bone tissue

Turning Bone into Bone: Ossification

There are three types of bone cells: *osteoblasts* (the builders), *osteoclasts* (the wreckers), and *osteocytes* (the maintenance workers). Osteocytes are mature bone cells, carrying out daily functions and influencing the other cells. The growth of bone tissue is carried out by osteoblasts through a process called *ossification*, of which there are two types: *endochondral* and *intramembranous*.

Endochondral ossification

During fetal development, most bones are first formed from hyaline cartilage through endochondral ossification. Over time, osteoblasts place calcium compounds (or other mineral salts) on the collagen fibers and remains of chondrocytes (cartilage cells). This *calcification* is the core of the ossification process. After birth, endochondral ossification begins in a secondary site — the epiphyseal plate. When *growth hormone* (GH) is released from the anterior pituitary, the chondrocytes in the plate are stimulated to divide and enlarge. This creates space within the bone that osteoblasts will calcify thus lengthening the bone. When osteoblasts are signaled to stop building (or because they've walled themselves off in a lacuna), they become osteocytes and they remain as such for the remainder of their lives.

Intramembranous ossification

Intramembranous ossification occurs during fetal development in many flat bones. They begin as a fibrous membrane and osteoblasts calcify the tissue from the center of the bone outward. As such, the edges of the skull bones are not yet calcified at birth; this allows for the baby's skull to compress, in a sense, during childbirth (which is also why many babies come out looking

like cone heads). The soft spot between the bones is called a *fontanel* and will continue ossification over the next year.

As development continues, flat, short, and irregular bones utilize this method to grow in size. Intramembranous growth occurring later in development like this is sometimes referred to as *appositional growth*. A thin layer of connective tissue (which can be cartilage but doesn't have to be) is built just beneath the periosteum and is then calcified by osteoblasts. The bone grows proportionally one layer, or membrane, at a time. In long bones, growth in width occurs in this manner, only the tissue is built outside of the endosteum instead.

In either process, osteoblasts will cease calcification once spongy bone has been formed. To build compact bone, new osteoblasts will be brought in to fill in the empty space starting near a blood vessel, building out one ring at a time (this is how a lamella is formed). Once your growth plate is closed, or fully calcified, you are unable to gain any height but your bones are never really done developing. They are constantly undergoing a process called *remodeling*. As we apply forces through exercise, or just our daily activities, the bone matrix weakens (picture a brand new brick wall compared to an old, weathered one). Osteoclasts are signaled to begin the remodeling process. They will secrete acid to dissolve the area's matrix. The calcium compounds are broken up and the calcium ions return to the bloodstream. After the osteoclasts clean up the area via phagocytosis, osteoblasts will move in and deposit new calcium compounds, thereby strengthening the bone.



REMEMBER Remodeling does not occur one bone at a time (like road construction) or even in large chunks. It is done everywhere, in little bits, all the time. This way, the structural integrity of the bone is never put in jeopardy.

Remodeling is also under hormonal control. Since the bones serve as a reservoir for minerals, calcium in particular, the body can trigger osteoblasts or osteoclasts depending on the need. When calcium in the blood is in abundance, the thyroid gland releases *calcitonin*, triggering the osteoblasts to create calcium compounds and deposit them into the matrix. When blood calcium levels are low, the parathyroid gland releases *parathyroid hormone*

(PTH) to signal the osteoclasts to break down matrix. Calcium ions are released into the bloodstream so normal functioning can continue; calcium ions are particularly important for muscle contraction as well as nervous communication.

41 What does calcitonin do?

- a. It regulates metabolism of calcium.
- b. It controls the development of Haversian systems.
- c. It influences the formation of bone marrow.
- d. It breaks down calcium compounds in the matrix.
- e. It encourages the proper layering of osteoclasts.

42-51 Fill in the blanks to complete the following sentences:

During fetal development, most bones are first laid down as **42.**_____ during the fifth week after conception. Development of the bone begins with **43.**_____, the depositing of mineral compounds. This is performed by **44.**_____ that attach themselves to the cartilage. Ossification in long bones begins in the **45.**_____ and moves toward the **46.**_____ of the bone. These areas remain separated by a layer of uncalcified cartilage called the **47.**_____.

Another bone cell that enters with the blood supply is the **48.**_____, which helps absorb the remnants of original cartilage as ossification occurs. Later it helps absorb bone tissue from the center of the long bone's shaft, forming the **49.**_____ cavity. Unlike bones in the rest of the body, those of the skull and mandible (lower jaw) are first laid down as **50.**_____. In the skull, the edges of the bone don't ossify in the fetus but remain membranous and form **51.**_____.

52 The constant breakdown and rebuilding of bone matrix is termed

- a. endochondral growth.
- b. primary calcification.
- c. ossification.

- d. intramembranous growth.
- e. remodeling.

53 Endochondral growth is triggered by which hormone?

- a. PTH
- b. Calcitonin
- c. Calcitriol
- d. GH
- e. Osteocalcin

54 Identify the differences between intramembranous and endochondral growth.

The Axial Skeleton: Keeping It All in Line

Just as the Earth rotates around its axis, the axial skeleton lies along the midline, or center, of the body. Think of your spinal column and the bones that connect directly to it — the rib (thoracic) cage and the skull. The tiny *hyoid* bone, which lies just above your larynx, or voice box, also is considered part of the axial skeleton, although it's the only bone in the entire body that doesn't articulate, or connect, with any other bone. It's also known as the *tongue bone* because the tongue's muscles attach to it. There are a total of 80 named bones in the axial skeleton, which supports the head and trunk of the body and serves as an anchor for the pelvic girdle.

Making a hard head harder

Of the 80 named bones in the axial skeleton, 29 are in (or very near) the skull. In addition to the hyoid bone, 8 bones form the cranium to house and protect the brain, 14 form the face, and 6 bones make it possible for you to hear.

Fortunately for the cramming student, most of the bones in the skull come in pairs. In the cranium there's just one of each of the following:

- » *Frontal* bone (forehead)
- » *Occipital* bone (back and base of the skull) containing occipital condyles, which articulate with the atlas of the vertebral column
- » *Ethmoid* bone (made of several plates, or sections, between the eye orbits in the nasal cavity)
- » *Sphenoid* bone (a butterfly-shaped structure that forms the floor of the cranial cavity and the backs of the orbits)

But there are two *temporal* bones each housing: (1) the hearing organs in the *auditory meatus*; (2) a styloid process, a long pointed process for anchoring muscles; (3) a mandibular fossa articulating with the condyle of mandible; and (4) a zygomatic process that projects anteriorly joining the zygomatic bone forming the cheek prominence. There are two *parietal* bones (roof and sides of the skull). These bones are attached along *sutures* called the following:

- » *Coronal* (located at the top of the skull between the two parietal bones)
- » *Squamosal* (located on the sides of the head surrounding the temporal bone)
- » *Sagittal* (along the midline atop the skull located between the two parietal bones)
- » *Lambdoidal* (forming an upside-down V — the shape of the Greek letter lambda — on the back of the skull)

In the face, there is only one *mandible* (jawbone) and one *vomer* dividing the nostrils, but there are two each of *maxillary* (upper jaw), *zygomatic* (cheekbone), *nasal*, *lacrimal* (a small bone in the eye socket), *palatine* (which makes up part of the eye socket, nasal cavity, and roof of the mouth), and *inferior nasal concha*, or turbinated, bones. The mandible has two lateral

condyles that articulate in the mandibular fossa of the temporal bone. (The maxillary bones are also called, simply, the *maxilla*.) Each ear contains three ossicles, or bonelets, which also happen to be the smallest bones in the human body: the *malleus*, *incus*, and *stapes*.

The floor of the cranial cavity contains several openings, or *foramina* (the singular is *foramen*), that allow various nerves and vessels to connect to the brain.

- » The holes in the ethmoid bone's *cribriform plate* are *olfactory foramina* that allow olfactory — or sense of smell — receptors to pass through to the brain. A process called the *crista galli* extends into the brain cavity for the attachment of the *meninges* (outer membranes) of the brain.
- » A large hole in the occipital bone called the *foramen magnum* allows the spinal cord to connect with the brain.
- » The sphenoid bone is riddled with foramina.
 - The *optic foramen* allows passage of the optic nerves.
 - The *jugular foramen* allows passage of the jugular vein and several cranial nerves.
 - The *foramen rotundum* allows passage of the *trigeminal nerve*, which is the chief sensory nerve to the face and controls the motor functions of chewing.
 - The *foramen ovale* allows passage of the nerves controlling the tongue, among other things.
 - The *foramen spinosum* allows passage of the middle *meningeal* artery, which supplies blood to various parts of the brain.
 - The sphenoid bone also features the *sella turcica*, or Turk's saddle, that cradles the pituitary gland and forms part of the *foramen lacerum*, which allows passage of several key components of the autonomic nervous system.

Encased within the frontal, sphenoid, ethmoid, and maxillary bones of the skull are several air-filled, mucous-lined cavities called *paranasal sinuses* (named for the bones in which they are contained). Although you may think their primary function is to drive you crazy with pressure and infections, the

sinuses actually lighten the skull's weight. They make it easier to hold your head up high; they warm and humidify inhaled air; and they act as resonance chambers to prolong and intensify the reverberations of your voice. These drain into the nose and cause so much trouble when you cry or have a cold. The temporal bone contains the *mastoid sinuses*, which drain into the middle ear (hence the earache referred to as *mastoiditis*).

Putting your backbones into it



REMEMBER The axial skeleton also consists of 33 bones in the vertebral column, laid out in four distinct curvatures, or areas.

- » **The cervical, or neck, curvature** has seven vertebrae, with the *atlas* and *axis* bones positioned in the first and second spots, respectively, forming the joint connecting the skull and vertebral column that allows you to turn your head. The axis, or second cervical vertebra, contains the *dens*, or *odontoid process*. The atlas rotates around the dens, turning the head. (In a sense, the atlas bone holds the world of the head on its shoulders, as the Greek god Atlas held the Earth.)
- » **The thoracic, or chest, curvature** has 12 vertebrae that articulate with the ribs, most of which attach to the sternum anteriorly by costal cartilage, forming the rib cage that protects the heart and lungs.
- » **The lumbar, or small of the back, curvature** contains five vertebrae and carries most of the weight of the body, which means that it generally suffers the most stress.
- » **The sacral, or pelvic, curvature** includes the five fused vertebrae of the *sacrum* anchoring the pelvic girdle by the *sacroiliac joint* and four fused vertebrae of the *coccyx*, or tailbone.



REMEMBER The spinal cord extends down the center of the vertebrae only from the base of the brain to the uppermost lumbar vertebrae.

Each vertebra, except the atlas, consists of a body and a vertebral arch, which

features a long dorsal projection called a *spinous process* that provides a point of attachment for muscles and ligaments. On either side of this are the *laminae*, broad plates of bone on the posterior surface that form a bony covering over the spinal canal. The laminae attach to the two *transverse processes*, which in turn are attached to the body of the vertebra by regions called the *pedicles*. The vertebrae align to form a large opening, called the *vertebral foramen*, allowing the passage of the spinal cord. Laterally, between the vertebrae, are openings called the *intervertebral foramina* that allow the spinal nerves to exit the vertebral column. The vertebra have superior articulating facets that articulate with inferior articulating facets of the adjacent vertebra, increasing the rigidity of the column, making the backbone more stable. Fibrocartilage discs located between the vertebrae act as shock absorbers. Openings in the transverse process, called the *transverse foramina*, allow large vessels and nerves ascending the neck to reach the brain.

Connecting to the vertebral column are the 12 pairs of ribs that make up the thoracic cage. All 12 pairs attach to the thoracic vertebrae, but the first 7 pairs attach to the *sternum*, or breastbone, by *costal cartilage*; they're called *true ribs*. Pairs 8, 9, and 10 attach to the cartilage of the seventh pair, which is why they're called *false ribs*. The last two pairs aren't attached in front at all, so they're called *floating ribs*.

The sternum has three parts:

- » **Manubrium:** The superior region that articulates with the clavicle, at the clavicular notch, and the first two pairs of ribs located up top, where you can feel a jugular notch in your chest in line with your *clavicles*, or collarbones.
- » **Body:** The middle part of the sternum forms the bulk of the breastbone and has notches on the sides where it articulates with the third through seventh pairs of ribs.
- » **Xiphoid process:** The lowest part of the sternum is an attachment point for the diaphragm and some abdominal muscles. (Interesting fact: Emergency medical technicians learn to administer CPR at least three finger widths above the xiphoid.)

55-67 Use the terms that follow to identify the bones, sutures, and landmarks of the skull shown in [Figure 7-3](#).

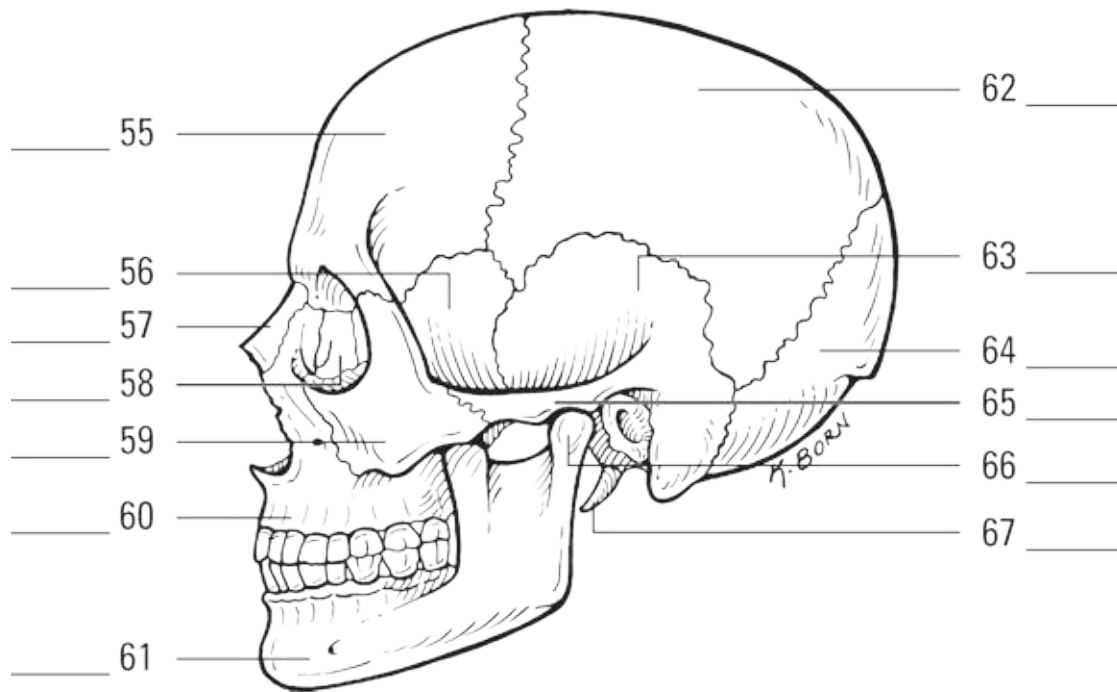


Illustration by Kathryn Born, MA

FIGURE 7-3: A lateral view of the skull.

- a. Temporal bone
- b. Sphenoid bone
- c. Nasal bone
- d. Styloid process
- e. Frontal bone
- f. Ethmoid bone
- g. Maxilla
- h. Parietal bone
- i. Occipital bone
- j. Zygomatic bone
- k. Mandibular condyle
- l. Zygomatic process
- m. Mandible

68-71 Match the skull bones with their connecting sutures.

- a. Squamosal suture
- b. Lambdoidal suture
- c. Sagittal suture
- d. Coronal suture

- 68. _____ Frontal and parietals
- 69. _____ Occipital and parietals
- 70. _____ Parietal and parietal
- 71. _____ Temporal and parietal

72-78 Use the terms that follow to identify the bones and landmarks of the skull shown in [Figure 7-4](#).

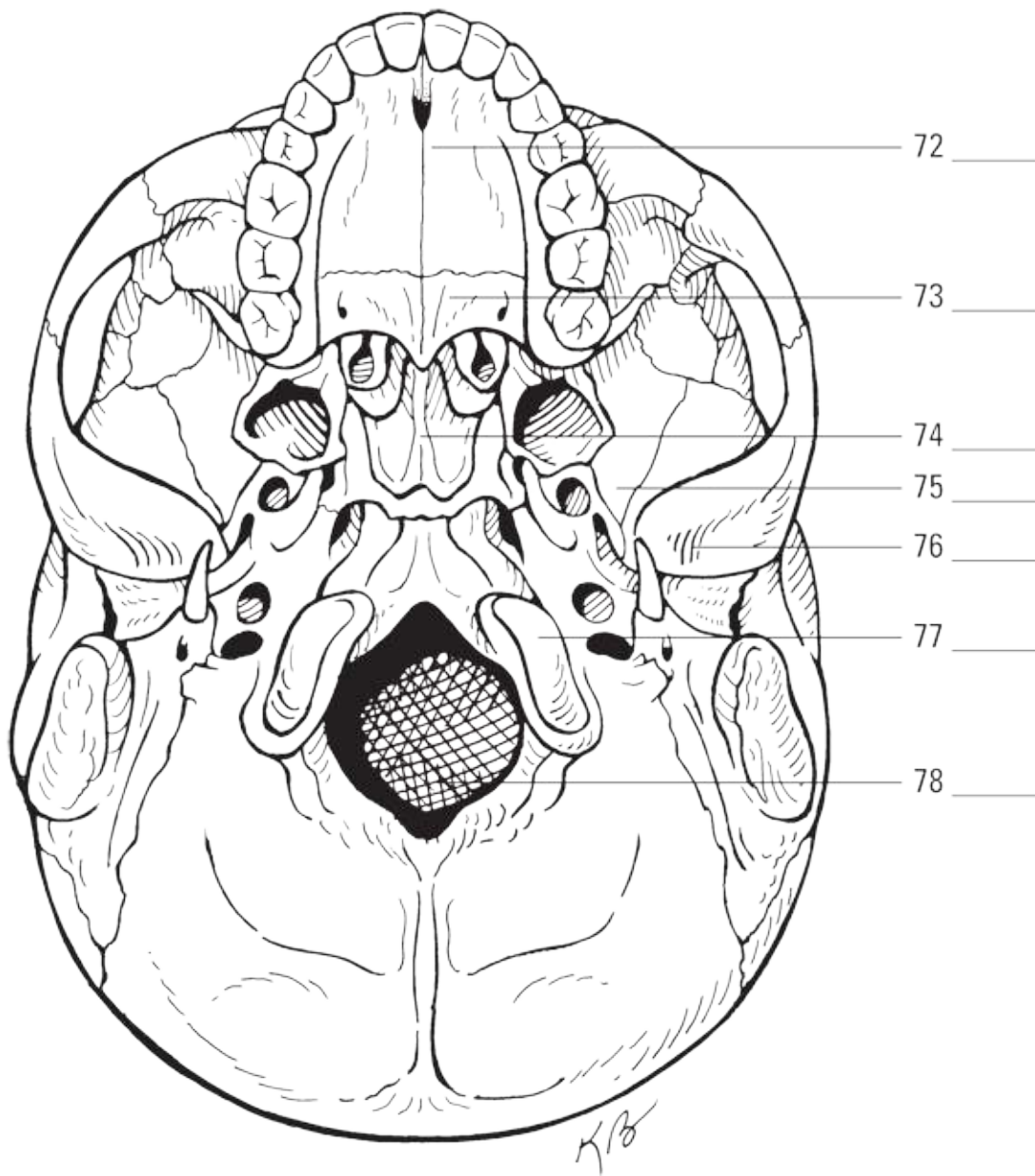


Illustration by Kathryn Born, MA

FIGURE 7-4: Inferior view of the skull.

- a. Vomer
- b. Mandibular fossa
- c. Foramen magnum
- d. Palatine bone
- e. Occipital condyle

f. Maxilla

g. Sphenoid bone

79-82 Use the terms that follow to identify the sinuses shown in [Figure 7-5](#).

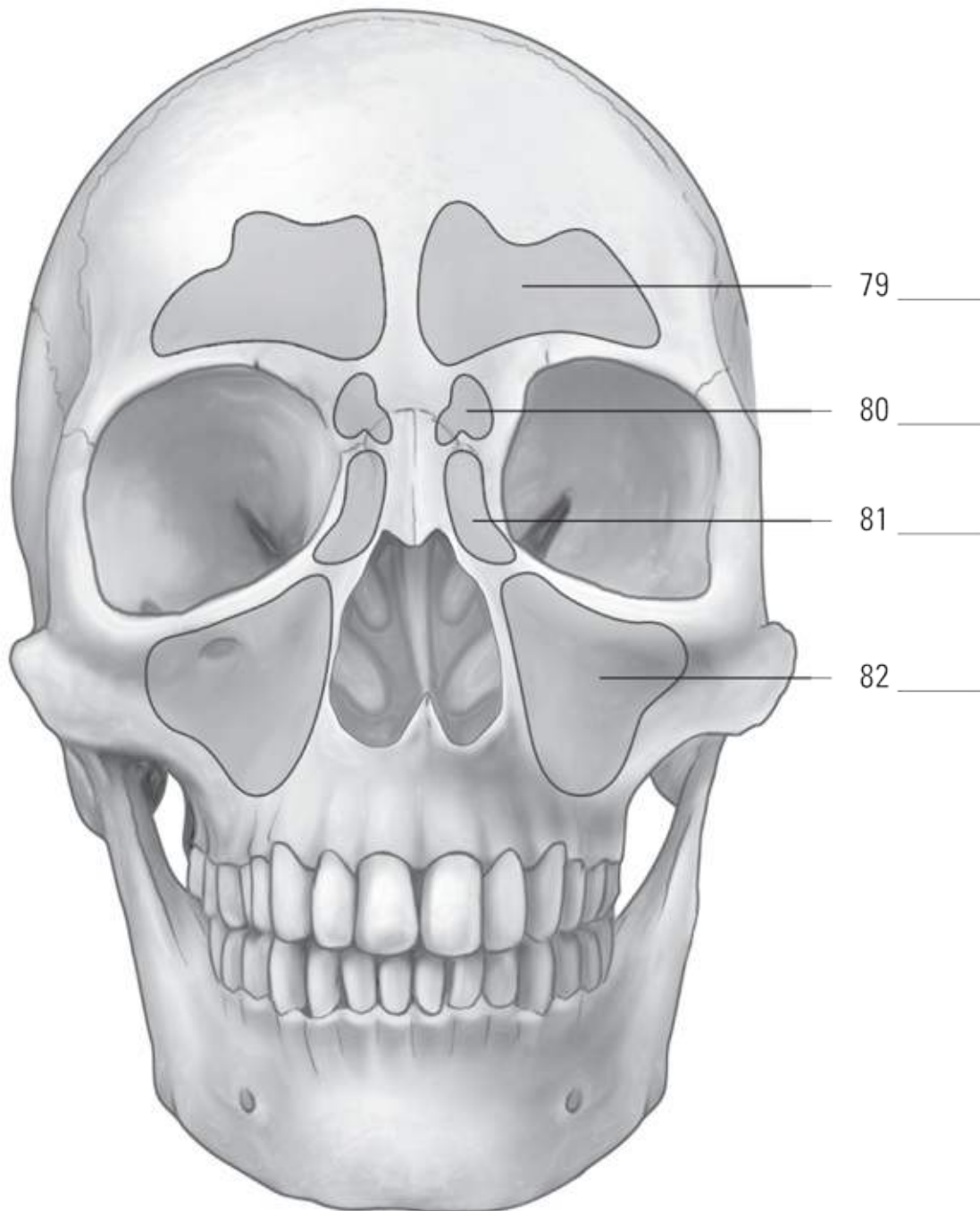


Illustration by Imagineering Media Services Inc.

FIGURE 7-5: Sinus view of the skull.

- a. Sphenoid sinus
- b. Frontal sinus
- c. Maxillary sinus
- d. Ethmoid sinus

83-91 Use the terms that follow to identify the regions, structures, and landmarks of the vertebral column shown in [Figure 7-6](#).

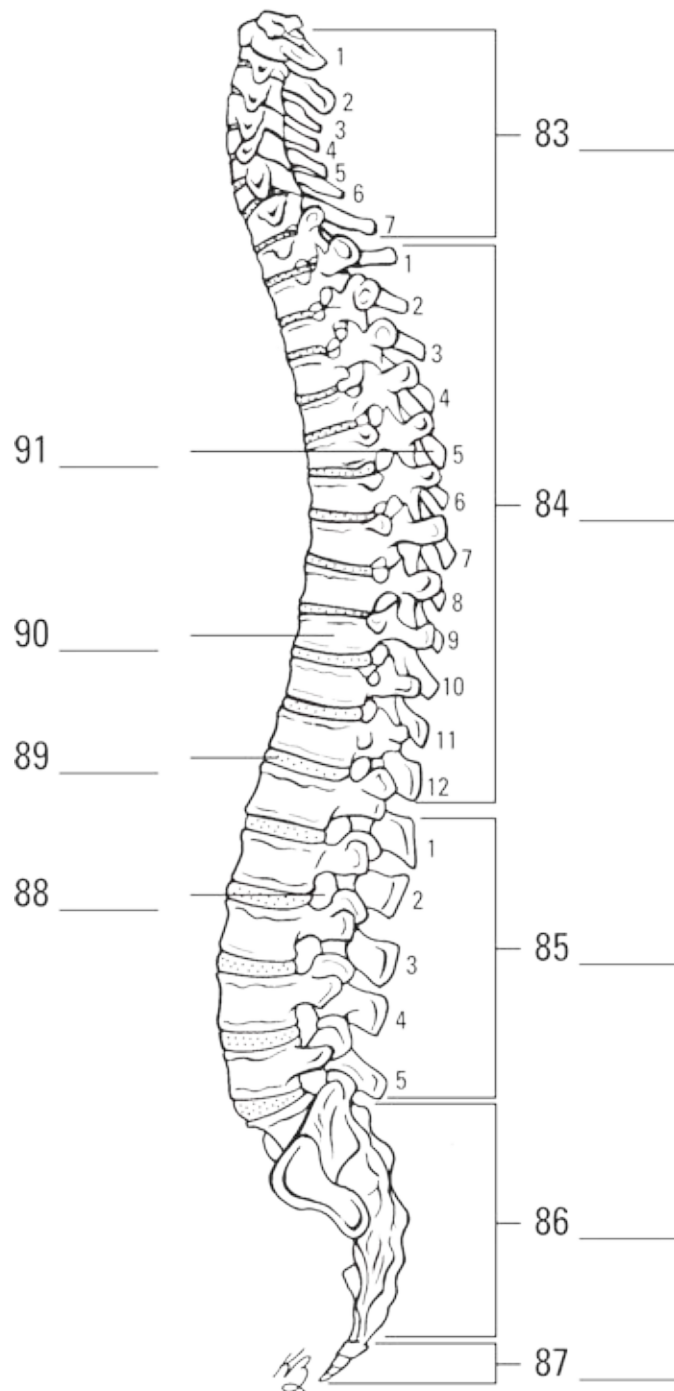


Illustration by Kathryn Born, MA

FIGURE 7-6: Vertebral column.

- Coccyx
- Intervertebral foramen
- Thoracic vertebrae or curvature

- d. Sacrum
- e. A vertebra
- f. Cervical vertebrae or curvature
- g. Lumbar vertebrae or curvature
- h. A spinous process
- i. Intervertebral disc

92-97 Fill in the blanks to complete the following sentences:

The organs protected by the thoracic cage include the **92**.

_____ and the **93**. _____. The first seven pairs of ribs attach to the sternum by the costal cartilage and are called **94**.

_____ ribs. Pairs 8 through 10 attach to the costal cartilage of the seventh pair and not directly to the sternum, so they're called **95**.

_____ ribs. The last two pairs, 11 and 12, are unattached anteriorly, so they're called **96**. _____ ribs. There's one bone in the entire skeleton that doesn't articulate with any other bones but nonetheless is considered part of the axial skeleton. It's called the **97**.

_____ bone.

The Appendicular Skeleton: Reaching beyond Our Girdles

Whereas the axial skeleton (described earlier in this chapter) lies along the body's central axis, the appendicular skeleton's 126 bones include those in all four appendages — arms and legs — plus the two primary girdles to which the appendages attach: the pectoral (chest) girdle and the pelvic (hip) girdle.

The pectoral girdle is made up of a pair of *clavicles*, or collarbones, which attach to the sternum medially and to the scapula laterally articulating with the *acromion process*, a bony prominence at the top of each of the pair of scapulae, better known as shoulder blades. Each scapula has a depression in it called the *glenoid fossa*, where the head of the *humerus* (upper arm bone) is attached. At the opposite end, the *capitulum* of the humerus articulates with the *trochlea* of the forearm's long *ulna* bone to form the elbow joint. The

process called the *olecranon* forms the elbow and is also referred to as the funny bone, although banging it into something usually feels anything but funny. The forearm also contains a bone called the *radius*. The biceps muscle attaches to the radial tuberosity, flexing the elbow. The radius, together with the ulna, articulates with the eight small *carpal* bones that form the wrist. The carpals articulate with the five *metacarpals* that form the hand, which in turn connect with the *phalanges* (finger bones), which are found as a pair in the thumb and as triplets in each of the fingers.

The pelvic girdle consists of two hipbones, called os *coxae*, as well as the *sacrum* and *coccyx*, more commonly referred to as the tailbone. During early developmental years, the os coxa consists of three separate bones — the *ilium*, the *ischium*, and the *pubis* — that later fuse into one bone sometime between the ages of 16 and 20. Posteriorly, the os coxa articulates with the sacrum, forming the *sacroiliac joint*, the source of much lower back pain; it's formed by the connection of the hipbones at the sacrum. Toward the front of the pelvic girdle, the two os coxae join to form the *symphysis pubis*, which is made up of fibrocartilage. A cuplike socket called the *acetabulum* articulates with the ball-shaped head of the *femur* (thigh bone). The femur is the longest bone in the body. The femur articulates with the *tibia* (shin bone) at the knee, which is covered by the *patella* (kneecap). Also inside each lower leg is the *fibula* bone, which joins with the tibia to connect with the seven *tarsal* bones that make up the ankle. These are the calcaneus (heel bone), talus, navicular, cuboid, lateral cuneiform, intermediate cuneiform, and medial cuneiform. The largest of these bones is the calcaneus. The tarsals join with the five *metatarsals* that form the foot, which in turn connect to the *phalanges* of the toes — a pair of phalanges in the big toe and triplets in each of the other toes.

98-114 Use the terms that follow to identify the bones and structures of the appendicular skeleton shown in [Figure 7-7](#).

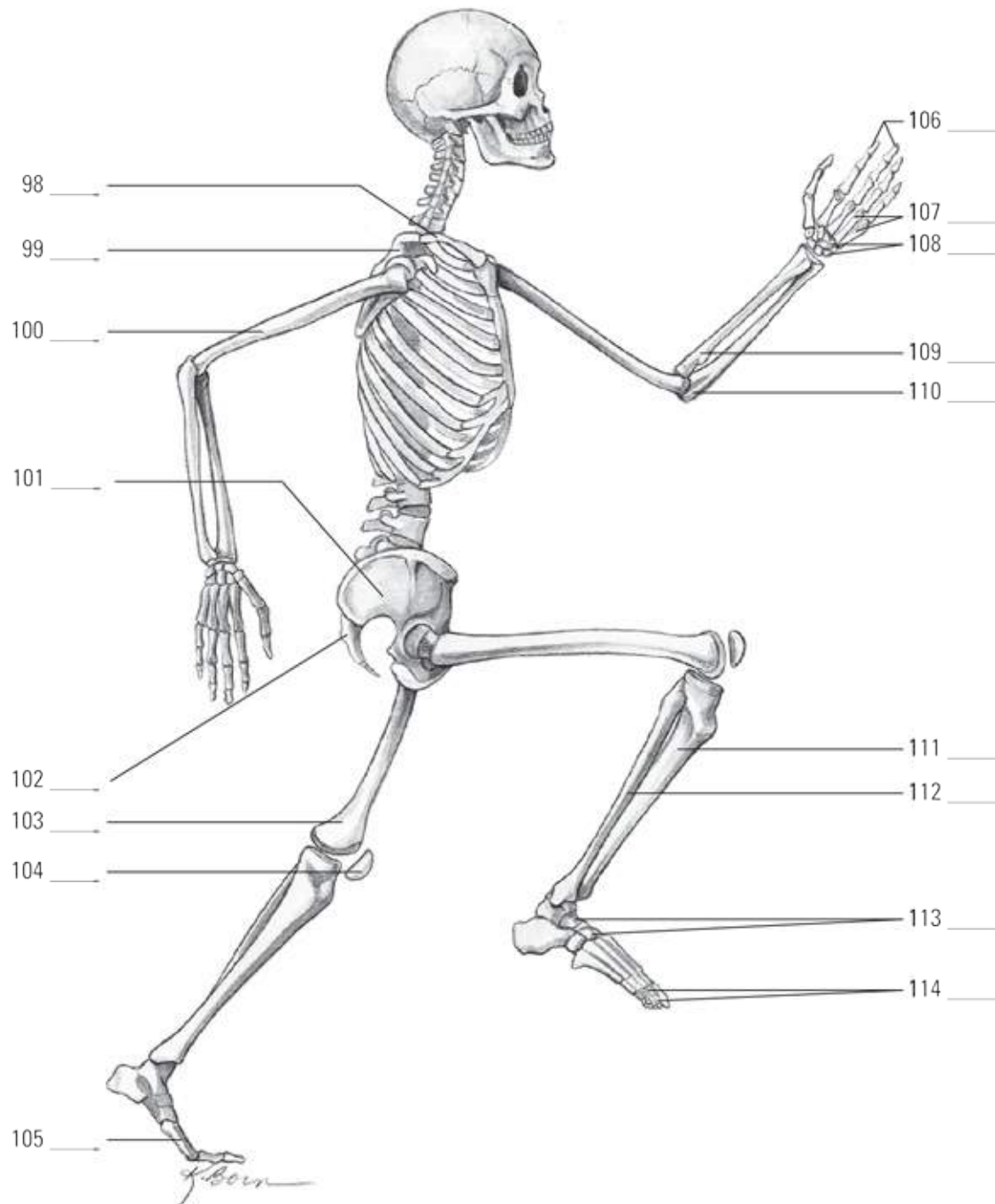


Illustration by Kathryn Born, MA

FIGURE 7-7: The appendicular skeleton.

- a. Tibia
- b. Ulna
- c. Scapula

- d. Metatarsals
- e. Carpals
- f. Phalanges of the foot
- g. Clavicle
- h. Fibula
- i. Patella
- j. Humerus
- k. Radius
- l. Os coxa
- m. Sacrum
- n. Phalanges of the hand
- o. Metacarpals
- p. Tarsals
- q. Femur

115 Where do the clavicles articulate with the scapulae?

- a. At the acromion process
- b. Below the glenoid fossa
- c. Behind the coracoid process
- d. At the acetabulum
- e. Along the upper spine

116 Where does the biceps muscle attach on the radius?

- a. The radial notch
- b. The humeral capitulum
- c. The ulnar trochlea
- d. The radial tuberosity
- e. The ulnar notch

117 Which of these bones IS NOT part of the pelvic girdle?

- a. Ilium
- b. Lumbar vertebrae
- c. Sacrum
- d. Ischium

118 What is the formal term for the prominence commonly referred to as the elbow?

- a. Olecranon process
- b. Trochlear notch
- c. Capitulum
- d. Radial notch
- e. Coronoid process

119 What is the formal term for the socket at the head of the femur?

- a. Obturator foramen
- b. Acetabulum
- c. Ischial tuberosity
- d. Symphysis pubis
- e. Greater sciatic notch

120 Which is largest and strongest tarsal bone?

- a. The talus
- b. The cuboid
- c. The cuneiform
- d. The calcaneus
- e. The metatarsal

121 Where are people who complain about their sacroiliac experiencing pain?

- a. Lower back
- b. Neck
- c. Feet
- d. Knees
- e. Hands

Arthrology: Articulating the Joints

Arthrology, which stems from the ancient Greek word *arthros* (meaning “jointed”), is the study of those structures that hold bones together, allowing them to move to varying degrees — or fixing them in place — depending on the design and function of the joint. The term *articulation*, or *joint*, applies to any union of bones, whether it moves freely or not at all. As such, joints can be classified either by structure or movement.



REMEMBER The movement classification of joints are as follows:

- » **Synarthrotic** — immovable
- » **Amphiarthrotic** — slightly movable (more like have “give” or allow compression)
- » **Diarthrotic** — highly movable

The three structural types of joints are as follows:

- » **Fibrous:** These are characterized by the fibrous connective tissue that joins the two bones. Most of these are synarthrotic and come in three forms:
 - *Sutures* occur when the bones are joined by a piece of dense connective tissue. These can be found between skull bones.
 - *Gomphosis* is found where the teeth attach to the sockets of the mandible and maxilla. They are joined by the *periodontal ligament*

(although teeth aren't actually bones).

- *Syndesmosis* is the only amphiarthrotic fibrous joint. Bones are joined by a sheet of connective tissue, like that of the articulations between the tibia and fibula. It allows for stretch and even some twisting.

» **Cartilaginous:** This type of joint is found in two forms:

- *Synchondrosis* is formed from rigid, hyaline cartilage that allows no movement. The most common example is the epiphyseal plate of the long bone. Other examples are the joint between the ribs, costal cartilage, and sternum.
- *Symphysis* occurs where fibrocartilage pads connect bones. Since the pads are flexible, these are amphiarthrotic (think about how you can arch your back). Examples include the intervertebral discs and the symphysis pubis.

» **Synovial:** All synovial joints are diarthrotic but to varying degrees. They are surrounded by a *joint capsule* that isolates the articulation leaving space that is filled with *synovial fluid* to reduce friction. Each adjoining bone is capped with hyaline cartilage called the *articular cartilage* and stabilized with ligaments. Some joints such as the knee and wrist have an additional pad of fibrocartilage called a *meniscus* between the bones for shock absorption. Many synovial joints also contain *bursae*, which are fluid filled sacs to reduce friction as tendons rub past the bones during movement.

There are six classifications of synovial joints:

- » **Gliding (or plane):** Curved or flat surfaces slide against one another, such as between the carpal bones in the wrist or between the tarsal bones in the ankle. Allows movement in two planes and rotation is possible but limited by their small size and ligament connections.
- » **Hinge:** A convex surface joints with a concave surface, allowing right-angle motions in one plane, such as elbows, knees, and joints between the phalanges.
- » **Pivot:** One bone pivots or rotates around a stationary bone, such as the atlas rotating around the dens of the axis at the top of the vertebral

column and the radioulnar joint. Rotation is the only type of movement allowed.

- » **Condylloid:** The oval head of one bone fits into a shallow depression in another, as in the carpal-metacarpal joint at the wrist and the tarsal-metatarsal joint at the ankle. Movement is allowed through all three planes but no rotation.
- » **Saddle:** Each of the adjoining bones is shaped like a saddle (the technical term is *reciprocally concavo-convex*), like in the carpometacarpal joint of the thumb (where the thumb meets the wrist). Movement is allowed through two planes and no rotation.
- » **Ball-and-socket:** The round head of one bone fits into a cuplike cavity in the other bone, allowing movement in all planes of motion plus rotation. These include your two most movable joints — the shoulder between the humerus and scapula and the hip between the femur and the os coxa.



REMEMBER The following are the main types of joint movement:

- » **Flexion:** A decrease in the angle between two bones
- » **Extension:** An increase in the angle between two bones
- » **Abduction:** Movement away from the midline of the body
- » **Adduction:** Movement toward the midline of the body
- » **Rotation:** Turning around an axis (medial or lateral)
- » **Pronation:** Downward or palm downward
- » **Supination:** Upward or palm upward
- » **Eversion:** Turning of the sole of the foot outward
- » **Inversion:** Turning of the sole of the foot inward
- » **Circumduction:** Forming a cone with the arm or leg (think tracing a circle)

22-127 Match the articulations with their joint types. Some joint types may be used more than once.

- a. Fibrous joint
- b. Cartilaginous joint-synchondrosis
- c. Cartilaginous joint-symphysis
- d. Synovial joint

122. _____ Sutures of the skull

123. _____ Between carpals

124. _____ Knee joint

125. _____ Symphysis pubis

126. _____ Epiphyseal plate

127. _____ Intervertebral discs

28-132 Use the terms that follow to identify the structures that form the synovial joint shown in [Figure 7-8](#).

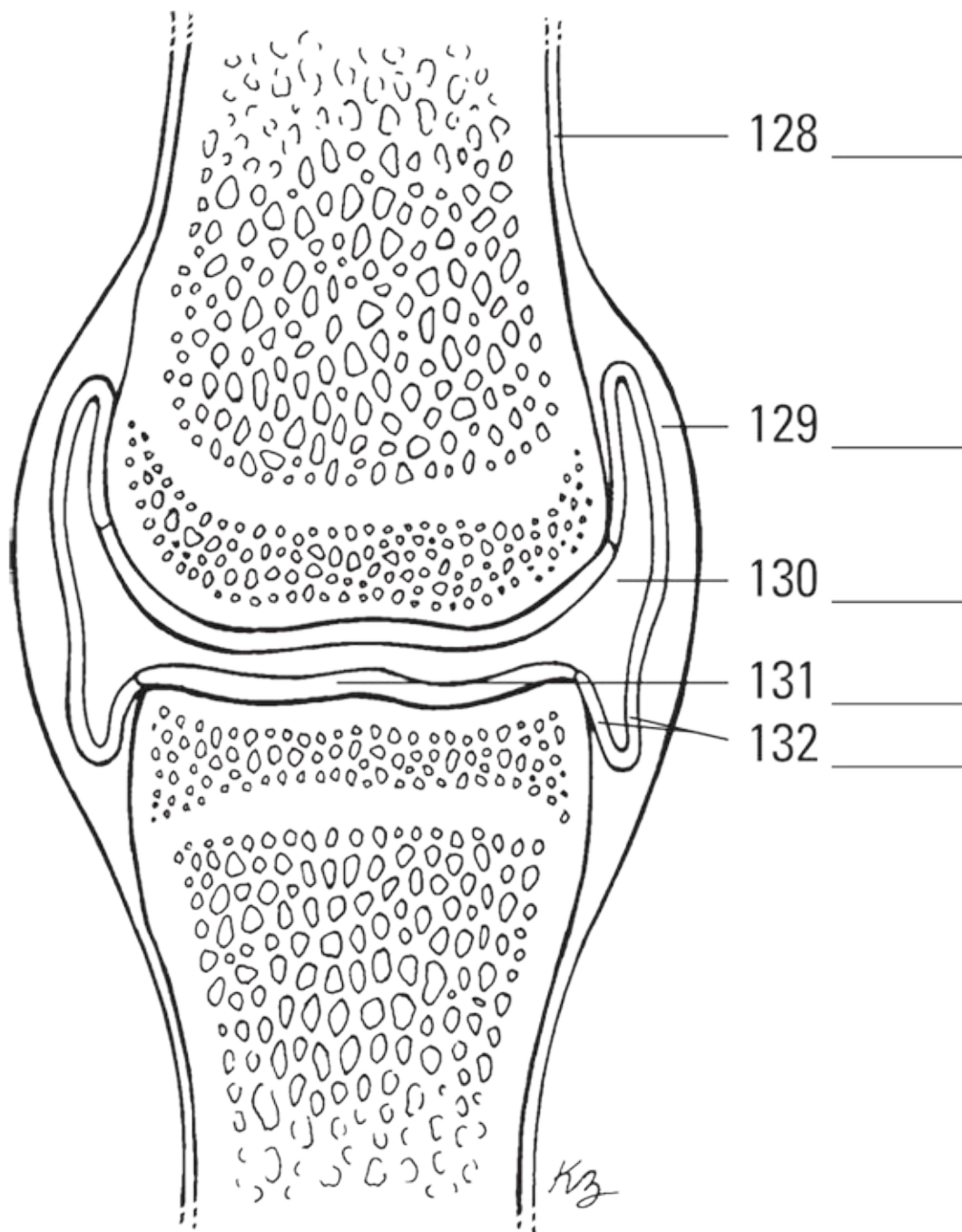


Illustration by Kathryn Born, MA

FIGURE 7-8: A synovial joint.

- a. Synovial (joint) cavity
- b. Periosteum
- c. Synovial membrane

- d. Articular cartilage
- e. Joint (fibrous) capsule

133 What is the formal term for an immovable joint?

- a. An amphiarthrosis
- b. A synarthrosis
- c. A syndesmosis
- d. A diarthrosis
- e. A synchondrosis

134 What material or structure allows free movement in a joint?

- a. The bursa
- b. The periosteum
- c. Synovial fluid
- d. The meniscus
- e. Bone marrow

135 Choose an example of a ball-and-socket joint.

- a. Between the finger and hand
- b. The hip
- c. The ankle
- d. The elbow
- e. Between the wrist bones

136 Choose an example of a pivot joint.

- a. Between the radius and the ulna
- b. Between the phalanges
- c. Between the mandible and the temporal bone

- d. Between the ribs and the sternum
- e. Between the tibia and the tarsals

137 What is the structure in the knee that divides the synovial joint into two separate compartments?

- a. The bursa
- b. Joint fat
- c. The tendon sheath
- d. The collateral ligaments
- e. The meniscus, or articular disc

38-147 Match the type of joint movement with its description.

- a. Upward or palm upward
- b. Decrease in the angle between two bones
- c. Turning of the sole of the foot inward
- d. Downward or palm downward
- e. Increase in the angle between two bones
- f. Turning of the sole of the foot outward
- g. Movement away from the midline of the body
- h. Forming a cone with the arm or leg
- i. Turning around an axis
- j. Movement toward the midline of the body

138. _____ Flexion

139. _____ Extension

140. _____ Abduction

141. _____ Adduction

142. _____ Rotation

143. _____ Pronation

- 144. _____ Supination
- 145. _____ Eversion
- 146. _____ Inversion
- 147. _____ Circumduction

Answers to Questions on the Skeleton

The following are answers to the practice questions presented in this chapter.

- ① What is happening during hematopoiesis? **b. Bone marrow is forming new blood cells.** Most of the other answers are nonsense.
- ② Which of the following IS NOT a role of minerals like phosphorous, calcium, and magnesium? **b. To provide cushioning and nourishment to the joints.** The minerals in the bone provide strength and bind together to organize the structure of the matrix, thus Choices a and d are both accurate. Bones also store the minerals — if too little is consumed, hormones can trigger the release of them from the bones.
- ③ In what way does the skeleton make locomotion possible? **b. Muscles use the skeleton as a scaffold against which they can contract.** You'd be a motionless blob without a skeleton to support coordinated movement, and, besides, the rest of the answers are utter nonsense.
- ④ Why do bones break more easily in older bodies than in younger ones? **b. Mineral content decreases, reducing their weight-bearing ability.** The correct answer becomes obvious through a process of elimination.
- ⑤ What does the periosteum do? **b. It allows bones to connect with ligaments and tendons.** Back to Greek again: *peri* means “around” and *osteon* means “bone,” so the periosteum is “around the bone.”
- ⑥ Where are Sharpey's fibers? **c. Between the periosteum and bone tissue.** Described by anatomist William Sharpey in 1846, these are also called *perforating fibers*. They connect the periosteum to the matrix of the bone tissue underneath.
- ⑦ What is inside the medullary cavity? **a. Bone marrow.** Usually you'll find

yellow marrow, although in infants red marrow also is present.

- ⑧ Volkmann's canals **a. are found in compact bone only**. Ironically, anatomist Alfred Wilhelm Volkmann was most noted for his observations of the physiology of the nervous system, not bones.
- ⑨ Which type of bones forms the weight-bearing part of the skeleton? **c. Long bones**. After all, they are the largest and most robust of the body's bones.
- ⑩ Vertebrae of the vertebral column: **b. Irregular bone**
- ⑪ Femur in thigh: **c. Long bone**
- ⑫ Sternum: **a. Flat bone**
- ⑬ Tarsals in ankle: **d. Short bone**
- ⑭ Humerus in upper arm: **c. Long bone**
- ⑮ Phalanges in fingers and toes: **c. Long bone**
- ⑯ Scapulae of shoulder: **a. Flat bone**
- ⑰ Kneecap: **d. Short bone** (or to be more specific, sesamoid)
- ⑱ Carpals in wrist: **d. Short bone**
- ⑲ An abrupt or pointed projection: **i. Spine**
- ⑳ A large, usually blunt process: **k. Trochanter**
- ㉑ A designation for any prominence or prolongation: **h. Process**
- ㉒ A large, often rough eminence: **m. Tuberosity**
- ㉓ A prominent ridge: **b. Crest**
- ㉔ A large, rounded articular end of a bone, often set off from the shaft by the neck: **f. Head**

- 25) An oval articular prominence of a bone: **a. Condyle**
- 26) A smooth, flat, or nearly flat articulating surface: **c. Facet**
- 27) A deeper depression: **e. Fossa**
- 28) A groove: **j. Sulcus**
- 29) A hole: **d. Foramen**
- 30) A canal or opening to a canal: **g. Meatus**
- 31) A smaller, rounded eminence: **i. Tubercle**

32-40) Following is how [Figure 7-2](#), the long bone, should be labeled.

32. **h. Articular cartilage**; 33. **g. Epiphyseal plate**; 34. **d. Spongy bone**;
35. **i. Compact bone tissue**; 36. **e. Medullary or nutrient artery**; 37. **b. Medullary cavity**; 38. **f. Proximal epiphysis**; 39. **a. Diaphysis**; 40. **c. Distal epiphysis**

- 41) What does calcitonin do? **a. It regulates metabolism of calcium.** Once you remember that calcitonin comes from the metabolism's pacemaker, the thyroid gland, you can discern the right answer.

42-51) Fill in the blanks to complete the following sentences:

Bones are first laid down as **42. cartilage** during the fifth week after conception. Development of the bone begins with **43. calcification**, the depositing of calcium and phosphorus. Next, the blood supply entering the cartilage brings **44. osteoblasts** that attach themselves to the cartilage. Ossification in long bones begins in the **45. diaphysis** of the long bone and moves toward the **46. epiphysis** of the bone. The epiphyseal and diaphyseal areas remain separated by a layer of uncalcified cartilage called the **47. epiphyseal plate**.

Another very large cell that enters with the blood supply is the **48. osteoclast**, which helps absorb the cartilage as ossification occurs. Later it helps absorb bone tissue from the center of the long bone's shaft, forming the **49. medullary or marrow** cavity. Unlike bones in the rest of the body, those of the skull and mandible (lower jaw) are first laid down as **50.**

membrane. In the skull, the edges of the bone don't ossify in the fetus but remain membranous and form **51. fontanelles.**

52) The constant breakdown and rebuilding of bone matrix is termed **e. remodeling.** Just think about remodeling a room of a house, you have to get rid of the stuff you don't want first.

53) Endochondral growth is triggered by which hormone? **d. GH.** Growth hormone (GH) triggers bone growth — easy to remember right? Calcitonin and calcitriol specifically trigger calcification by osteoblasts, and PTH (parathyroid hormone) is the only hormone that triggers osteoclasts. Osteocalcin is a structural protein, not a hormone.

54) Identify the differences between intramembranous and endochondral growth. **These two processes share more in common than differences. Once osteoblasts are triggered to deposit mineral compounds, the steps are the same. The difference lies in how they create the space. Endochondral growth uses hyaline cartilage as a template while intramembranous uses fibrous connective tissue. In fetal development, all bones are formed via endochondral ossification with the exception of the flat bones of the skull and the mandible (which utilize endochondral). Throughout life, only long bones use endochondral growth (to increase their length). All other bones use intramembranous growth to increase in size proportionally, one layer at a time. Long bones do this to increase their width.**

55-67) Following is how [Figure 7-3](#), the lateral view of the skull, should be labeled.

55. **e. Frontal bone;** 56. **b. Sphenoid bone;** 57. **c. Nasal bone;** 58. **f. Ethmoid bone;** 59. **j. Zygomatic bone;** 60. **g. Maxilla;** 61. **m. Mandible;** 62. **h. Parietal bone;** 63. **a. Temporal bone;** 64. **i. Occipital bone;** 65. **l. Zygomatic process;** 66. **k. Mandibular condyle;** 67. **d. Styloid process**

68) Frontal and parietals: **d. Coronal suture**

69) Occipital and parietals: **b. Lambdoidal suture**

70) Parietal and parietal: **c. Sagittal suture**

(71) Temporal and parietal: **a. Squamosal suture**

(72-78) Following is how [Figure 7-4](#), the inferior view of the skull, should be labeled.

72. **f. Maxilla**; 73. **d. Palatine bone**; 74. **a. Vomer**; 75. **g. Sphenoid bone**;
76. **b. Mandibular fossa**; 77. **e. Occipital condyle**; 78. **c. Foramen magnum**

(79-82) Following is how [Figure 7-5](#), the sinus view of the skull, should be labeled.

79. **b. Frontal sinus**; 80. **d. Ethmoid sinus**; 81. **a. Sphenoid sinus**; 82. **c. Maxillary sinus**

(83-91) Following is how [Figure 7-6](#), the vertebral column, should be labeled.

83. **f. Cervical vertebrae or curvature**; 84. **c. Thoracic vertebrae or curvature**; 85. **g. Lumbar vertebrae or curvature**; 86. **d. Sacrum**; 87. **a. Coccyx**; 88. **b. Intervertebral foramen**; 89. **i. Intervertebral disc**; 90. **e. A vertebra**; 91. **h. A spinous process**

(92-97) Fill in the blanks to complete the following sentences:

The organs protected by the thoracic cage include the **92. heart** and the **93. lungs**. The first seven pairs of ribs attach to the sternum by the costal cartilage and are called **94. true** ribs. Pairs 8 through 10 attach to the costal cartilage of the seventh pair and not directly to the sternum, so they're called **95. false** ribs. The last two pairs, 11 and 12, are unattached anteriorly, so they're called **96. floating** ribs. There's one bone in the entire skeleton that doesn't articulate with any other bones but nonetheless is considered part of the axial skeleton. It's called the **97. hyoid** bone.

(98-114) Following is how [Figure 7-7](#), the frontal view of the skeleton, should be labeled.

98. **g. Clavicle**; 99. **c. Scapula**; 100. **j. Humerus**; 101. **l. Os coxa**; 102. **m. Sacrum**; 103. **q. Femur**; 104. **i. Patella**; 105. **d. Metatarsals**; 106. **n. Phalanges of the hand**; 107. **o. Metacarpals**; 108. **e. Carpals**; 109. **k. Radius**; 110. **b. Ulna**; 111. **a. Tibia**; 112. **h. Fibula**; 113. **p. Tarsals**; 114. **f. Phalanges of the foot**

- 115) Where do the clavicles articulate with the scapulae? **a. At the acromion process**
- 116) Where does the biceps muscle attach on the radius? **d. The radial tuberosity**
- 117) Which of these bones is not part of the pelvic girdle? **b. Lumbar vertebrae**
- 118) What is the formal term for the prominence commonly referred to as the elbow? **a. Olecranon process**
- 119) What is the formal term for the socket at the head of the femur? **b. Acetabulum**
- 120) Which is the largest and strongest tarsal bone? **d. The calcaneus**
- 121) Where are people who complain about their sacroiliac experiencing pain? **a. Lower back**
- 122) Sutures of the skull: **a. Fibrous joint**
- 123) Between carpals: **d. Synovial joint**
- 124) Knee joint: **d. Synovial joint**
- 125) Symphysis pubis: **c. Cartilaginous joint-symphysis**
- 126) Epiphyseal plate: **b. Cartilaginous joint-synchondrosis**
- 127) Intervertebral discs: **c. Cartilaginous joint-symphysis**
- 128-132) Following is how [Figure 7-8](#), the synovial joint, should be labeled.
128. **b. Periosteum**; 129. **e. Fibrous capsule**; 130. **a. Synovial (joint) cavity**; 131. **d. Articular cartilage**; 132. **c. Synovial membrane**
- 133) What is the formal term for an immovable joint? **b. A synarthrosis**
- 134) What material or structure allows free movement in a joint? **c. Synovial fluid**
- 135) Choose an example of a ball-and-socket joint. **b. The hip**

- 136) Choose an example of a pivotal joint. **a. Between the radius and the ulna**
- 137) What is the term for the structure in the knee that divides the synovial joint into two separate compartments? **d. Meniscus or articular disc**
- 138) Flexion: **b. Decrease in the angle between two bones**
- 139) Extension: **e. Increase in the angle between two bones**
- 140) Abduction: **g. Movement away from the midline of the body**
- 141) Adduction: **j. Movement toward the midline of the body**
- 142) Rotation: **i. Turning around an axis**
- 143) Pronation: **d. Downward or palm downward**
- 144) Supination: **a. Upward or palm upward**
- 145) Eversion: **f. Turning of the sole of the foot outward**
- 146) Inversion: **c. Turning of the sole of the foot inward**
- 147) Circumduction: **h. Forming of a cone with the arm or leg**

Chapter 8

Getting in Gear: The Muscles

IN THIS CHAPTER

- » Understanding the functions and structure of muscles
 - » Classifying types of muscle
 - » Getting the scoop on muscles as organs
 - » Breaking down muscle contractions, tone, and power
 - » Deciphering muscle names
-

Much of what we think of as “the body” centers around our muscles and what they can do, what we want them to do, and how tired we get trying to make them do it. With all that muscles do and are, it’s hard to believe the word “muscle” is rooted in the Latin word *musculus*, which is a diminutive of the word for “mouse.” Well, the muscle is a mouse that roars. Muscles make up most of the fleshy parts of the body and account for 30-50 percent of the body’s weight (depending on sex with females being on the lower end). Layered over the skeleton (which we discuss in [Chapter 7](#)), they largely determine the body’s form. More than 500 muscles are large enough to be seen by the unaided eye, and thousands more are visible only through a microscope.

There are three distinct types of muscle tissue, classified by their differing structure and control, but they share the following characteristics:

- » **Contractility:** Ability to shorten
- » **Extensibility:** Ability to stretch
- » **Elasticity:** Ability to return to resting length
- » **Excitability:** Ability to respond to stimulus
- » **Conductivity:** Ability to create an electrical signal

Flexing Your Muscle Knowledge



REMEMBER The study of muscles is called *myology* after the Greek word *mys*, which also means “mouse.” Muscles perform a number of functions vital to maintaining life, including

- » **Movement:** Skeletal muscles (those attached to bones) convert chemical energy into mechanical work, producing movement ranging from finger tapping to a swift kick of a ball by contracting, or shortening. Reflex muscle reactions protect your fingers when you put them too close to a fire and startle you into watchfulness when an unexpected noise sounds. Many purposeful movements require several sets, or groups, of muscles to work in unison.
- » **Vital functions:** Without muscle activity, you die. Muscles are doing their job when your heart beats, when your blood vessels constrict, and when your intestines squeeze food along your digestive tract in *peristalsis*.
- » **Antigravity:** Perhaps that’s overstating it, but muscles do make it possible for you to stand and move about in spite of gravity’s ceaseless pull. Did your mother tell you to improve your posture? Just think how bad it would be without any muscles!
- » **Heat generation:** You shiver when you’re cold and stamp your feet and jog in place when you need to warm up. That’s because chemical reactions in muscles create heat as a byproduct, helping to maintain the body’s temperature.
- » **Joint stability:** Muscles and their tendons aid the ligaments to reinforce joints.
- » **Supporting and protecting soft tissues:** The body wall and floor of the pelvic cavity support the internal organs and protect soft tissues from injury.
- » **Sphincters:** These muscles encircle openings to control swallowing, defecation, and urination.

As you may remember from studying tissues (see [Chapter 5](#) for details), muscle cells — called *fibers* — are some of the longest in the body. Some muscle fibers contract rapidly, whereas others move at a leisurely pace. Generally speaking, however, the smaller the structure to be moved, the faster the muscle action. Exercise can increase the thickness of muscle fibers, but it doesn't make new fibers. Skeletal muscles have a rich vascular supply that dilates during exercise to give the working muscle the extra oxygen it needs to keep going.

Two processes are central to muscle development in the developing embryo: *myogenesis*, during which muscle tissue is formed, and *morphogenesis*, when the muscles form into internal organs. By the eighth week of gestation, a fetus is capable of coordinated movement.



REMEMBER Following are some important muscle terms to know:

- » **Fascia:** Layers of dense connective tissue found between muscles; maintains a muscle's position relative to the other muscles around it
- » **Fiber:** An individual muscle cell
- » **Insertion:** The more movable attachment of a muscle
- » **Origin:** The immovable attachment of a muscle, or the point at which a muscle is anchored by a tendon to the bone
- » **Motor neuron:** A neuron that stimulates contraction of a muscle
- » **Myofibril:** Fibrils (fiber-like structures) within a muscle cell that contain the protein filaments actin and myosin that slide during contraction, shortening the fiber (or cell)
- » **Striated:** A characteristic of cardiac and skeletal muscle tissues; span the width of the cell
- » **Sarcoplasm:** The cellular cytoplasm in a muscle fiber; similarly the *sarcolemma* is the cell membrane
- » **Tendon:** Dense regular connective tissue made up of collagen, a fibrous protein that attaches muscles to bone; allows some muscles to apply their force at some distance from where a contraction actually takes place

- » **Aponeurosis:** Broad sheet of connective tissue that attaches muscles to other muscles (or sometimes bones)
- » **Tone, or tonus:** State of tension present to a degree at all times, even when the muscle is at rest

Complete the following practice questions to see how well you understand the basics of myology:

1 Which of the following is NOT a true statement?

- a. Muscle fibers are some of the longest cells in the body.
- b. Myofibrils within muscle cells contain protein filaments that slide during contraction.
- c. Sphincters hold fibers together to form muscles.
- d. Muscles create actions that both move the body and move things through it.
- e. Posture is an expression of muscle action.

2 What happens during myogenesis and morphogenesis?

- a. Muscles that aren't flexed regularly begin to lose tone during myogenesis and are destroyed entirely during morphogenesis.
- b. Collagen is converted into muscle tissue during myogenesis and recycled by the body during morphogenesis.
- c. Chemical energy is converted into mechanical work during myogenesis and morphogenesis.
- d. Embryonic muscles form during myogenesis, and muscles form into internal organs during morphogenesis.

3 Why do your muscles shiver when you're cold?

- a. To prevent paralysis
- b. To generate heat
- c. To improve elasticity
- d. To increase excitability

e. To maintain contractility

4 Name the cellular unit in muscle tissue.

- a. Filament
- b. Myofibril
- c. Fiber
- d. Motor cell
- e. Fasciculus

5 What defines a state of tension present to a degree at all times?

- a. Rigor
- b. Tonus
- c. Clovus
- d. Excitability
- e. Paralysis

6 It's possible to completely relax every muscle in the body.

- a. True
- b. False

7 Exercise forms new muscle fibers.

- a. True
- b. False

Muscle Classifications: Smooth, Cardiac, and Skeletal



REMEMBER Muscle tissue is classified in three ways based on the tissue's function, shape, and structure:

- » **Smooth muscle tissue:** So-called because these spindle-shaped fibers don't have the striations typical of other kinds of muscle. This muscle tissue forms into sheets and makes up the walls of hollow organs such as the stomach, intestines, and bladder. The tissue's involuntary movements are relatively slow, so contractions last longer than those of other muscle tissue, and fatigue is rare. If arranged in a circle inside an organ, contraction constricts the cavity inside. If arranged lengthwise, contraction of smooth muscle tissue shortens the organ.
- » **Cardiac muscle tissue:** Found only in the heart, cardiac muscle fibers are cylindrical, striated, feature one central nucleus, and move through involuntary control. An electron microscope view of the tissue shows separate fibers tightly pressed against each other, forming cellular junctions called *intercalated discs* that look like tiny, dark-colored plates. These intercalated discs are gap junctions that help move an electrical impulse throughout the heart. For more on cardiac muscle, flip to [Chapter 11](#).
- » **Skeletal muscle tissue:** This is the tissue that most people think of as muscle. It's the only muscle that can be voluntarily controlled via the somatic nervous system (SoNS). The long, striated, cylindrical fibers of skeletal muscle tissue contract quickly but tire just as fast. Skeletal muscle, which is also what's considered meat in animals, is 20 percent protein, 75 percent water, and 5 percent organic and inorganic materials. Each multinucleated fiber is encased in a thin, transparent cell membrane called a *sarcolemma* that receives and conducts stimuli. Packed into the cells are numerous tiny myofibrils, roughly 1 micron in diameter, that are suspended in the cell's sarcoplasm. Some skeletal muscle fibers have *muscle spindles* wrapped around them. These specialty sensory cells function in *proprioception*. They communicate with the brain to relay information on muscle actions which is translated into a sense of body position.

The following practice questions test your knowledge of muscle classifications:

8 What type of muscle tissue lacks striations?

- a. Cardiac
- b. Smooth
- c. Skeletal
- d. Fascia
- e. Contracting

9 What does a sarcolemma do?

- a. It receives and conducts stimuli for skeletal muscle tissue.
- b. It provides a structure for moving electrical impulses through the heart.
- c. It forms a wall for hollow organs.
- d. It is made up of the filaments responsible for contraction.
- e. It prevents muscle fatigue.

10 Which muscle type appears only in a single organ?

- a. Contractile
- b. Smooth
- c. Cardiac
- d. Skeletal
- e. Fascia

11 Intercalated discs

- a. anchor skeletal muscle fibers to one another.
- b. play a role in moving electrical impulses through the heart.
- c. are found only in the muscles of the back.
- d. overlap to cause contraction.

e. contribute to tactile perception.

12-14 Identify the muscle tissue illustrated in [Figure 8-1](#)

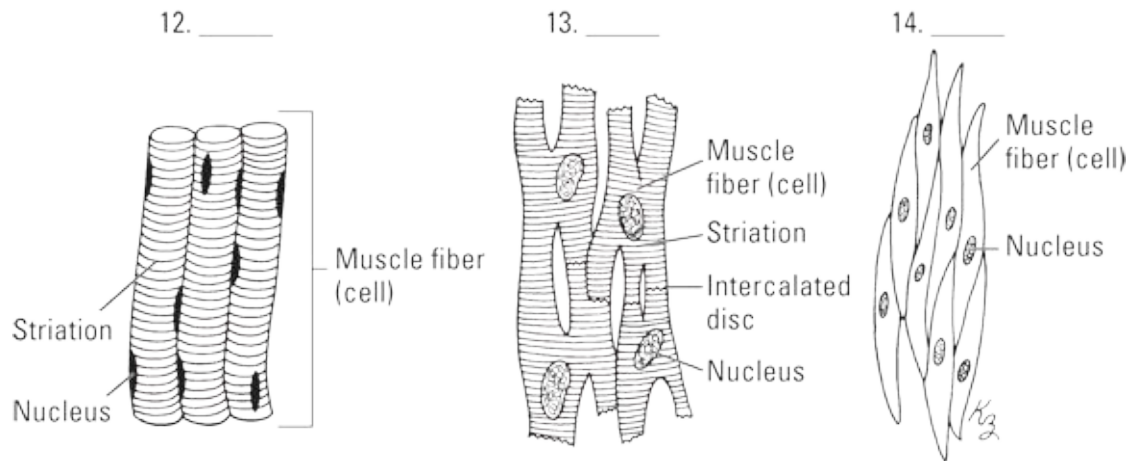


Illustration by Kathryn Born, MA

FIGURE 8-1: Muscle tissue types.

- a. skeletal
- b. smooth
- c. cardiac

Contracting for a Contraction

Before we can explain how and why muscles do what they do, it's important that you understand the anatomy of how they're put together. Use [Figure 8-2](#) as a visual guide as you read through the following sections.

Structure	Description	Contents
Z-line	Thin line down the middle of the light band (I)	The protein that holds actin filaments in position, also attaches to myosin via titin
M-line	Thin line down the middle of the dark band (A)	Proteins that keep myosin filaments in line with each other
A-band	Dark block that creates striations	Myosin and any other proteins found within its length such as actin and titin
H-zone	Lighter block in the center of the dark band	Area of myosin that does not have actin overlapping it
I-band	Light band that creates striations	Actin filaments that don't overlap with myosin, the Z lines, and titin

For the muscle to contract, the sarcomere must shorten; that is, the Z lines get pulled closer together. This is accomplished by the interaction between actin and myosin. As the filaments slide past each other, the H zone is reduced or obliterated, pulling the Z lines closer together and reducing the I bands. (The A bands don't change.) Voilà! Contraction has occurred! This is explained in the sliding filament theory, which is illustrated in [Figure 8-3](#).

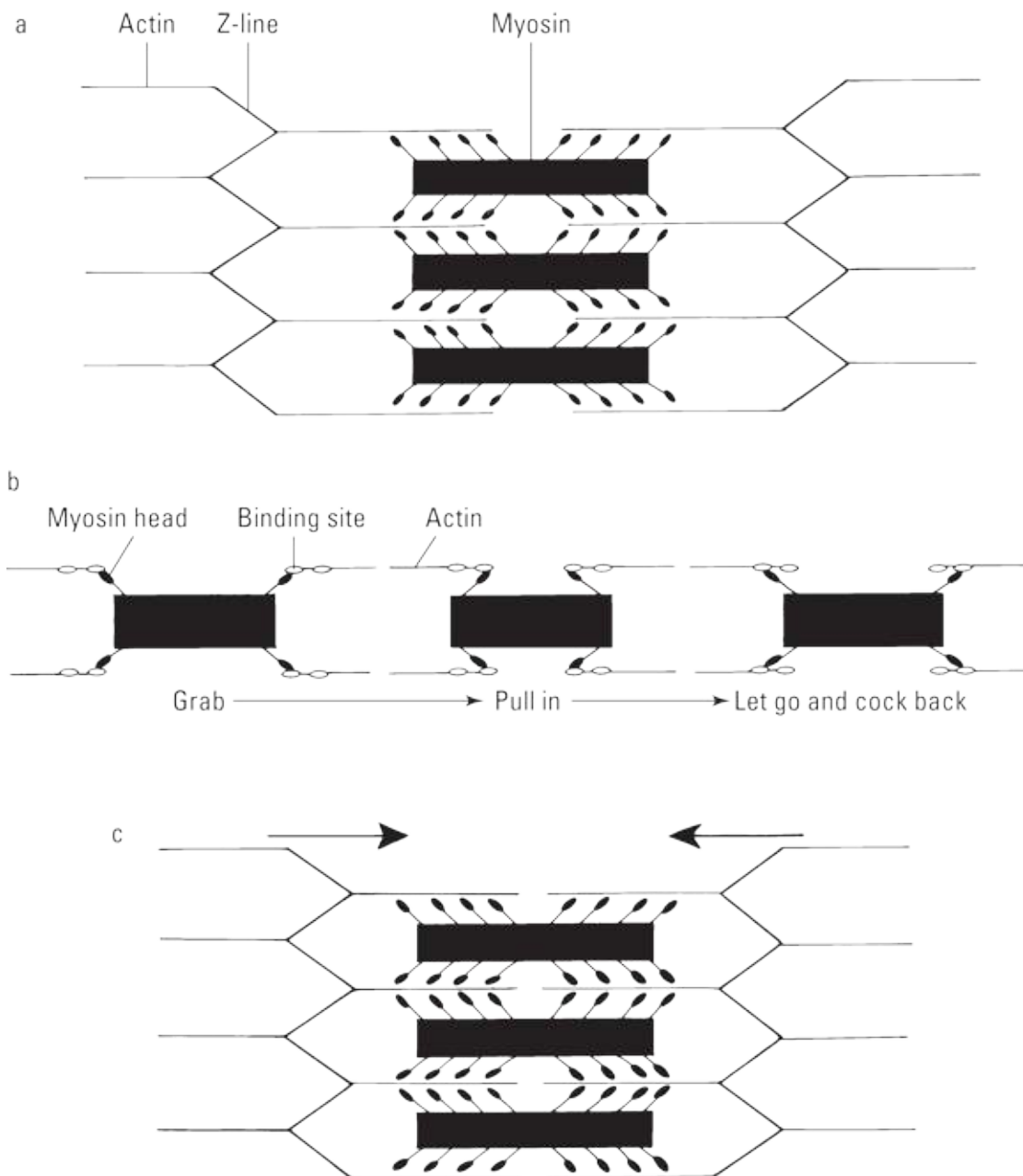


Illustration by Kathryn Born, MA

FIGURE 8-3: Sarcomere structure and shortening: a) sarcomere before contraction; b) close-up view of a power stroke; c) contracted sarcomere showing Z lines closer together.

The myosin filaments contain numerous heads that point away from the center of the sarcomere (see [Figure 8-3a](#)). The actin filaments contain binding sites for these myosin heads. Contraction begins when these heads grab onto the actin, forming a *cross bridge*, and perform *power strokes* (see [Figure 8-3b](#)). The first part of a power stroke requires no energy — cross bridges form and the myosin heads reflexively pull in. This alone, however, does not

generate enough force to move a body part. To finish a power stroke, ATP comes in and binds to the myosin head, causing it to let go of the actin. Then, ATP is catalyzed and the energy is used to cock the heads back into their original position. (For a review of ATP, refer to [Chapter 2](#).) The myosin heads are now able to perform another power stroke, this time grabbing a binding site further down on the actin filament. As long as the muscle is being stimulated and ATP is available, the *cross bridge cycling* can continue.

Because it requires no energy to form a cross bridge between a myosin head and the actin, the binding sites must remain covered until contraction is triggered by the associated motor neuron. This is achieved through a pair of proteins known as the *t-t complex*. *Tropomyosin* wraps around the actin filament, covering up the binding sites. *Troponin* pins the tropomyosin to the actin, ensuring it doesn't slip off. The final step of stimulation uncovers the binding sites.

Understanding what stimulates muscle contraction

After you know how muscles contract, you need to figure out what stimulates them to do so. We cover the details of the nervous system in [Chapter 9](#), but here you can find out what's happening as an impulse stimulates a skeletal muscle.

The *stimulus*, from the central nervous system, is brought to the muscle through an *efferent* (outgoing) nerve to a motor neuron which will directly stimulate the fiber. Each individual muscle fiber must be stimulated by a motor neuron and has a designated place along its sarcolemma for this, the *motor end plate*. Fortunately, a single motor neuron has numerous branches that allow it to trigger multiple fibers at once, creating motor units. *Large motor units* are able to generate great force, controlling our gross motor skills (such as walking). *Small motor units* trigger only a few fibers so they control our fine motor skills (such as writing) — sacrificing force to gain more intricate control.

A motor neuron communicates with a fiber using a neurotransmitter called *acetylcholine* (ACh). Once the neuron releases ACh, it moves across the space between the neuron and the muscle fiber which is called the *synaptic cleft*. ACh then binds to a receptor on the motor end plate. This triggers the fiber to generate an *impulse*. This is the same process neurons use in nervous communication, which is described in detail in [Chapter 9](#). For now, picture a

spark moving down a power line like in cartoons.

The impulse created by the fiber is quickly spread around the cell through the sarcolemma (cell membrane). It then spreads deep into the cell via structures called *t-tubules*. These carry the impulse to the *sarcoplasmic reticulum (SR)*. The SR wraps around the myofibrils and stores calcium ions. When stimulated by impulse, the SR releases the calcium ions. The ions will then bind to troponin, shifting the tropomyosin off the binding sites on actin. Myosin heads can now grab onto the actin, initiating the first power stroke.



REMEMBER Does it matter whether the signal received is strong or weak? Nope. That's the *all-or-none law* of muscle contraction. The fiber either contracts completely or not at all. In other words, if a single muscle fiber is going to contract, it's going to do so to its fullest extent.

Following are some practice questions that deal with muscle anatomy and contraction:

15 Which of these terms doesn't belong in the following list?

- a. Anisotropic
- b. Actin
- c. Myosin
- d. Sarcolemma
- e. Troponin

16 During contraction, what does NOT occur?

- a. Actin and myosin filaments shorten.
- b. Thick and thin filaments slide past each other.
- c. Muscle fibers shorten.
- d. Myosin heads pull on actin filaments.
- e. Troponin binds with calcium ions.

17 A weak stimulus causes a muscle fiber to contract only partway.

- a. True
- b. False

18 Which of the following steps of muscle contraction requires the breakdown of ATP?

- a. release of calcium ions
- b. uncovering of binding sites
- c. myosin heads pulling actin closer
- d. cross bridges breaking off from the binding sites
- e. myosin heads cocking back to their original position

19-23 Label the sarcomere illustrated in [Figure 8-4](#)

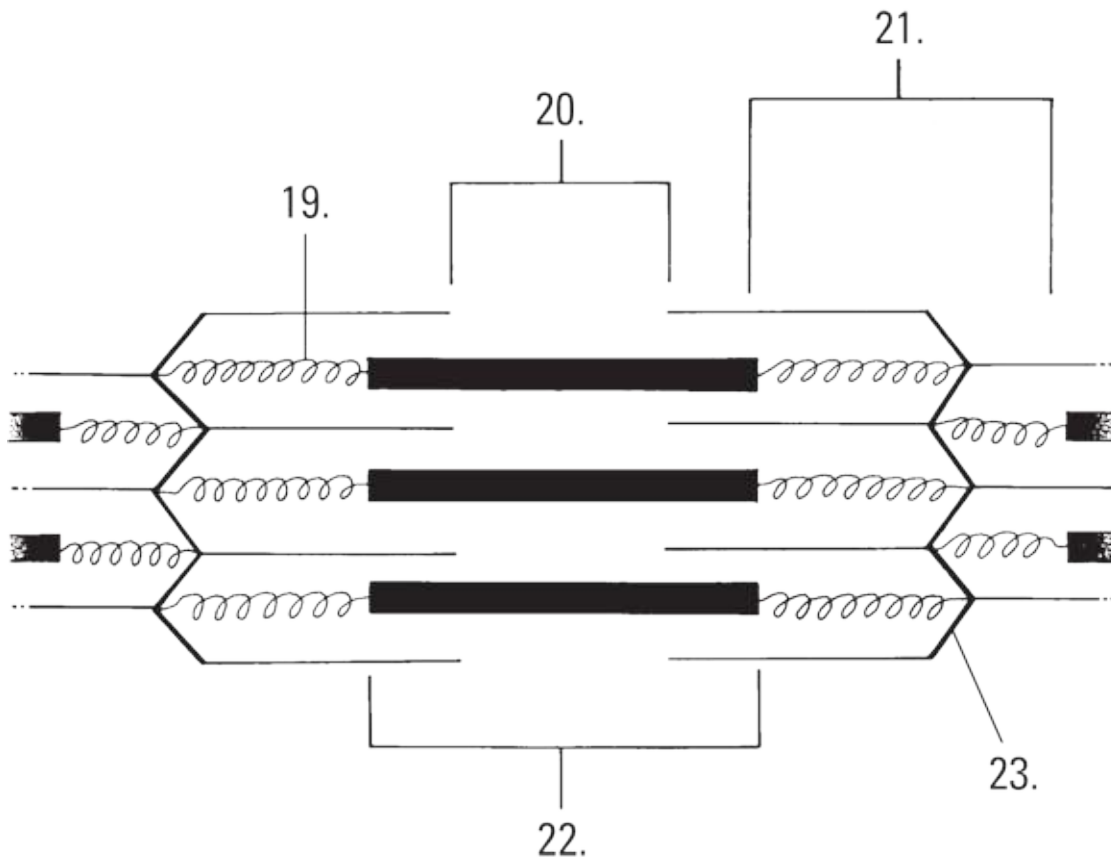


Illustration by Kathryn Born, MA

FIGURE 8-4: Sarcomere.

- a. Titin
- b. Z lines
- c. I band
- d. A band
- e. H zone

24 Which statement best characterizes a large motor unit?

- a. Many neurons stimulate the same muscle, generating a large force.
- b. One neuron stimulates all the fibers of a muscle for precise control.
- c. It controls precise movements such as writing.
- d. Many neurons stimulate many fibers for a powerful contraction.
- e. One neuron stimulates many fibers for a powerful movement.

25 Outline the steps of contraction stimulus, ending with exposure of the binding sites.

Pulling Together: Muscles as Organs



REMEMBER A muscle organ has two parts:

» **The belly:** Composed predominantly of muscle fibers.

» **The tendon:** Composed of fibrous, or collagenous, regular connective tissue. If the tendon is a flat, sheetlike structure attaching a wide muscle, it's called an *aponeurosis*.

The sarcolemma of each muscle fiber is surrounded by areolar connective tissue called *endomysium* that binds the fibers together into bundles called *fascicles* (see [Figure 8-5](#)). Each bundle is surrounded by areolar connective tissue called *perimysium*. All the fascicles together make up the belly of the muscle, which is surrounded by dense connective tissue called the *epimysium*. Blood vessels, lymphatic vessels, and nerves can be found wrapped in the epimysium as well as between the fascicles.

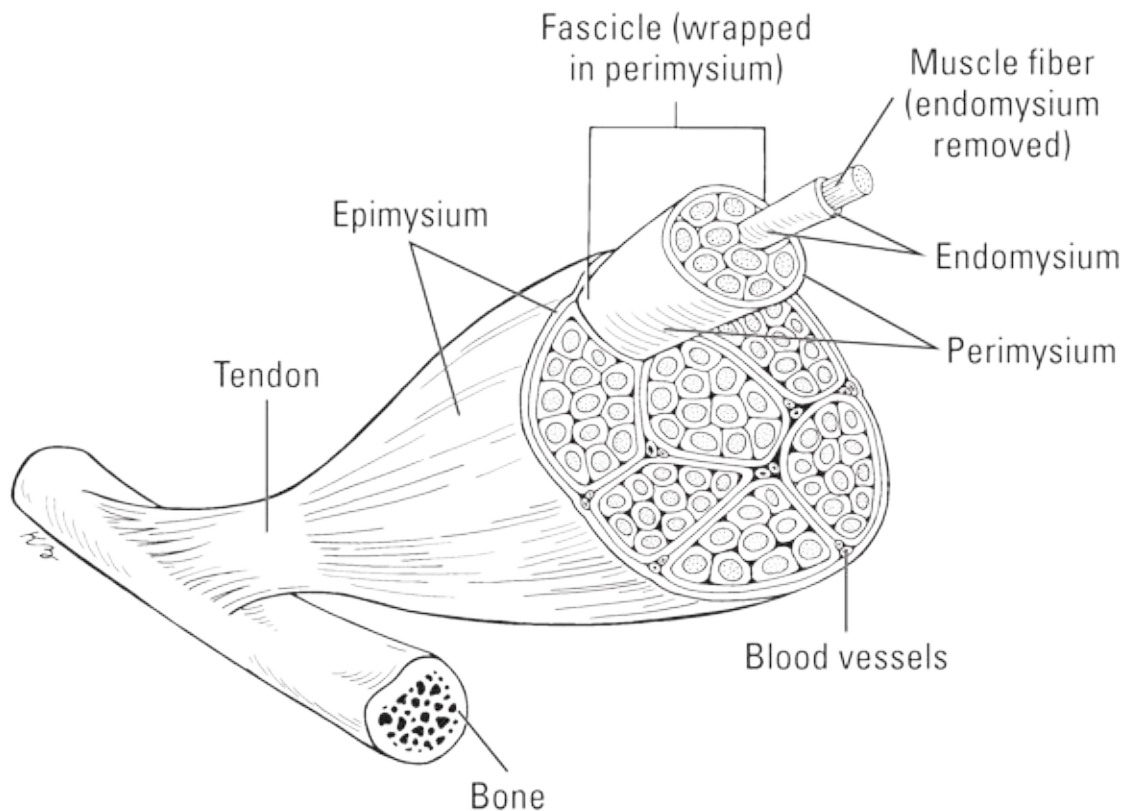


Illustration by Kathryn Born, MA

FIGURE 8-5: Organizational structure of a muscle.

When actin and myosin overlap, the sarcomeres shorten pulling the ends of the myofibrils closer together. Since the myofibrils span the length of the fiber, the fiber shortens as a result. Similarly, the fibers run about the length of the muscle so the entire belly shortens, pulling on the tendon.

Cardiac and smooth muscle contract in the same manner but are organized

into their organs differently. Cardiac muscle makes up the majority of the heart's walls. The cells, however, connect end-to-end, often to multiple cells (giving it its branched appearance). The myofibrils are still packed in parallel within the fibers so it is also striated in appearance. Cardiac muscle fibers usually have only one, centrally located nucleus. Smooth muscle is the least organized of the three. Like cardiac fibers, it has a single, central nucleus. The myofibrils, however, are not bundled in parallel, they crisscross the cell. As a result, smooth muscle is not striated, and instead of contracting by pulling ends together, it pulls in from all directions. This is why smooth muscle is found in the walls of hollow organs.

26-30 Match the muscle structures with their descriptions.

- a. Perimysium
- b. Aponeurosis
- c. Epimysium
- d. Fascicles
- e. Endomysium

- 26. _____ Areolar tissue that surrounds fibers
- 27. _____ Bundles of muscle fibers
- 28. _____ Connective tissue that wraps muscle fibers into bundles
- 29. _____ Connective tissue that creates a functional muscle
- 30. _____ Flat, sheet-like tendon that serves as insertion for a large, flat muscle

31 Which of the following muscle tissues are striated?

- I. skeletal
 - II. cardiac
 - III. smooth
- a. I only
 - b. III only
 - c. I and III

- d. I and II
- e. all of the above

32 While structurally different, all three muscle fiber types contract their myofibrils utilizing the sliding-filament model.

- a. True
- b. False

Assuming the Right Tone

As we note in the earlier section “[Understanding what stimulates muscle contraction](#),” when it comes to contraction of a muscle fiber, it’s an all-or-nothing affair. Bear in mind, though, that this applies to a fiber, not the muscle as a whole. While the actin begins to slide back once the myosin heads let go, if the fiber is being continuously stimulated, they won’t reach their resting position. When pulled in again, they overlap even more, increasing the force generated with each power stroke. This is referred to as *summation*. If stimulus lasts long enough, there will be no relaxation between each contraction cycle, which is known as *tetanic contraction*.

Multiple neurons forming multiple motor units allow us to contract only a portion of a muscle, or the whole thing at once. More so, some motor units require greater stimulation than others which allows us to control the force of contraction. Fibers that are easier to stimulate are triggered first. As the stimulus intensifies, more motor units are activated, which is referred to as *recruitment*. Recruitment ensures that muscle movements are smooth rather than jerky as the fibers *summate* to sustained contraction. Even when a muscle is at rest, some of its fibers maintain a level of sustained contraction called *tone*, or *tonus*. Without this, we wouldn’t be able to stand or even hold our heads up!



REMEMBER In physiology, a muscle contraction is referred to as a *muscle twitch*. A twitch is the fundamental unit of recordable muscular activity. The twitch consists of a single stimulus-contraction-relaxation sequence in a

muscle fiber. The contraction is the period when the cross bridges are cycling. The relaxation phase occurs when cross bridges detach and the muscle tension decreases. Complete fatigue occurs when no more twitches can be elicited, even with increasing intensity of stimulation.



REMEMBER There are two distinct types of muscle contraction:

- » **Isometric:** Occurs when a contracting muscle is unable to move a load, for example, trying to push a building to one side or doing a wall sit. It retains its original length but develops *tension*. No mechanical work is accomplished, and all energy involved is expended as heat. The muscles that control your posture maintain isometric contractions.
- » **Isotonic:** Occurs when the resistance offered by the load (or the gardening hoe or the cold can of soda) is less than the tension developed, thus shortening the muscle and resulting in mechanical work. There are two main types of isotonic contraction:
 - **Concentric contraction:** Occurs when the force generated by the muscle is less than the maximum tension. The muscle shortens with increased velocity. An example is the upward motion of a biceps curl (elbow flexion).
 - **Eccentric contraction:** Occurs when the force of a load is greater than the maximum muscle tension, causing the muscle to elongate while performing work. This often occurs in conjunction with a concentric contraction on the opposite side. For example, the downward motion of a biceps curl: While the triceps perform the concentric contraction, the controlled motion demands an eccentric contraction from the biceps.

As such, muscles aren't independent, sole proprietors. Each muscle depends upon companions in a muscle group to assist in executing a particular movement. That's why muscles are categorized by their actions. The brain coordinates the following groups through the cerebellum.

- » **Prime movers:** Provide the major force for producing a specific movement. Just as it sounds, these muscles are the workhorses that

produce movement.

- » **Antagonists:** These muscles exist in opposition to prime movers, regulating the motion by contracting and providing resistance.
- » **Fixators or fixation muscles:** These muscles serve to steady a part while other muscles execute movement. They don't actually take part in the movement itself.
- » **Synergists:** These muscles control movement of the proximal joints so that the prime movers can bring about movements of distal joints.

Flex your knowledge of muscle tone and function with these practice questions:



EXAMPLE Q. Muscle contraction that is unable to move an object involves an action known as

- a. isometric.
- b. eccentric.
- c. isotonic.
- d. concentric.

A. The correct answer is isometric. When the tension leads to movement (actual work), it's isotonic. Eccentric contraction and concentric contraction are the two types of isotonic contraction.

33 What do you call muscles that tend to counteract or slow an action?

- a. Antagonists
- b. Fixators
- c. Primary movers
- d. Agonists
- e. Synergists

34 What creates a strong muscle contraction?

- a. Isotonic influences building up as mechanical work
- b. Isometric influences building up as tension
- c. Low frequency stimulatory nerve impulses
- d. High frequency stimulatory nerve impulses

35-38 Match the terms with their definitions.

- a. Summation
 - b. Tone
 - c. Recruitment
 - d. Tetanus
35. _____ As stimulation increases, more motor units are triggered to contract.
36. _____ When twitches occur before the previous one ends, generating more force.
37. _____ Stimulation occurs so frequently that no relaxation occurs at all.
38. _____ Constant, sustained contraction of some fibers within a muscle.

Leveraging Muscular Power

Skeletal muscle power is nothing without lever action. The bone acts as a rigid bar, the joint is the fulcrum, and the muscle applies the force. Levers are divided into the *weight arm*, the area between the fulcrum and the weight; and the *power arm*, the area between the fulcrum and the force. When the power arm is longer than the weight arm, less force is required to lift the weight, but range, or distance, and speed are sacrificed. When the weight arm is longer than the power arm, the range of action and speed increase, but power is sacrificed. Therefore, 90 degrees is the optimum angle for a muscle to attach to a bone and apply the greatest force.



REMEMBER Three classes of levers are at work in the body:

- » **First-class, or seesaw:** The fulcrum is located between the weight and the force being applied. An example is a nod of the head: The head-neck joint is the fulcrum, the head is the weight, and the muscles in the back of the neck apply the force.
- » **Second-class, or wheelbarrow:** The weight is located between the fulcrum and the point at which the force is applied. An example is standing on your tiptoes: The fulcrum is the joint between the toes and the foot, the weight is the body, and the muscles in the back of the leg at the heel bone apply the force.
- » **Third class, or removing a nail with a hammer:** The force is located between the weight and the fulcrum. An example is flexing your arm and showing off your biceps: The elbow joint is the fulcrum, the weight is the lower arm and hand, and the biceps insertion on the lower arm applies the force.

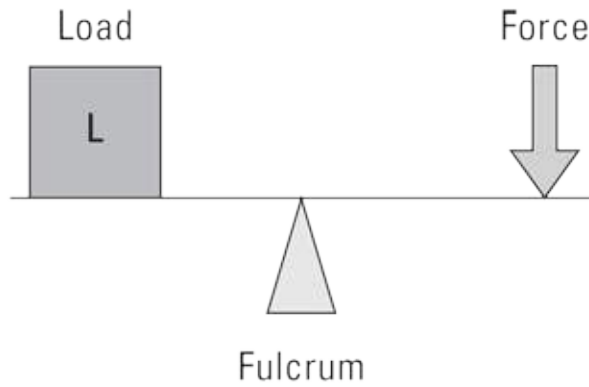
The power behind muscle contraction comes from the breakdown of ATP, which is generally made via cellular respiration. There are fibers, though, that can create ATP through lactic acid fermentation (see [Figure 2-11](#) in [Chapter 2](#)). These fibers, referred to as *fast-twitch*, do not need oxygen and can thus respond very quickly, generating great force. However, this anaerobic process generates only two ATP per glucose whereas aerobic respiration produces 38. So these fast-twitch fibers cannot maintain contraction to near as long as their *slow-twitch* counterparts. Slow-twitch fibers are thus used for endurance, but they don't contain as many myofibrils, so they can't generate as much force. These fibers need room for *myoglobin*, which stores oxygen for respiration. All of our muscles contain both fiber types as well as intermediate fibers (which fall between slow- and fast-twitch in their characteristics). We can't rely on slow-twitch alone because they cycle more slowly and don't generate as much force. And we can't rely on fast-twitch alone because they tire out quickly, and the fermentation creates *lactic acid* as a byproduct.



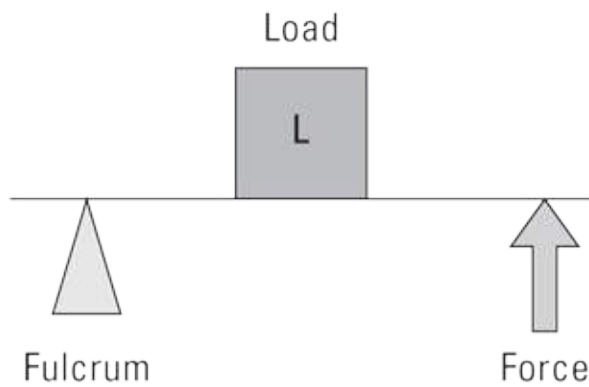
WARNING When lactic acid builds up, fibers can *fatigue*: They become unable to contract. This is distinct from a *cramp*, which is a sustained, involuntary contraction. While cramps and fatigue can occur simultaneously within a muscle, cramps come about from an electrolyte

imbalance in the fluid surrounding the cell, not the accumulation of lactic acid. Got all that? Then try your hand at the following questions:

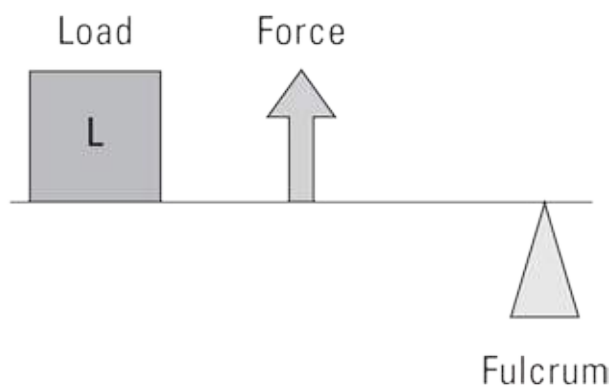
39 Which of the following in [Figure 8-6](#) is a Class II lever?



a



b



c

© John Wiley & Sons, Inc.

FIGURE 8-6: The three classes of muscle levers.

40 Which muscles would provide the force in a Class III lever?

- a. Biceps brachii
- b. Pectoralis major
- c. Orbicularis oculi
- d. Rectus abdominus
- e. Sartorius

41 Which of the following would produce a wide range of movement with speed while sacrificing power?

- a. Power arm and weight arm of equal lengths
- b. Long weight arm, short power arm
- c. Long power arm, shorter weight arm
- d. Short power arm, shorter weight arm

What's in a Name? Identifying Muscles



REMEMBER It may seem like a jumble of meaningless Latin at first, but muscle names follow a strict convention that names them according to one or more of the following:

- » **Function:** These muscle names usually have a verb root and end in a suffix (*-or* or *-eus*), followed by the name of the affected structure. Example: levator scapulae (elevates the scapulae).
- » **Compounding points of attachment:** These muscle names blend the origin and insertion attachment with an adjective suffix (*-eus* or *-is*). Examples: sternocleidomastoideus (sternum, clavicle, and mastoid process) and sternohyoideus (sternum and hyoid).
- » **Shape or position:** These muscle names usually have descriptive

adjectives that may be followed by the names of the locations of the muscles. Examples: rectus (straight) femoris, rectus abdominis, and serratus (sawtooth) anterior.

» **Figurative resemblance:** These muscle names are based on the muscles' resemblance to some objects. Examples: gastrocnemius (resembles the stomach) and trapezius (resembles a trapezoid or kite shape).

Check out [Figure 8-7](#) and [Table 8-2](#) for a rundown of prominent muscles in the body and key points to remember about each one.

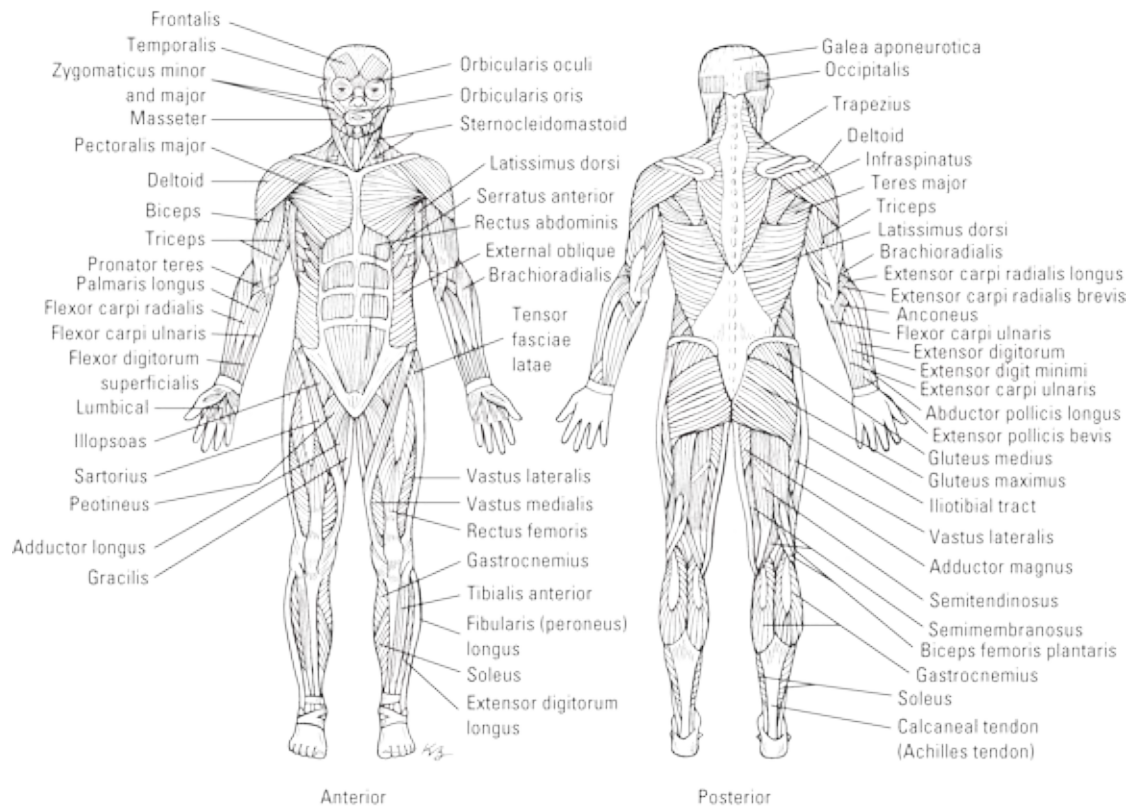


Illustration by Kathryn Born, MA

FIGURE 8-7: Some of the body's major muscles.

Table 8-2 Muscles of the Body

Muscle	Origin	Insertion	Action
Head			
Frontalis	Galea aponeurotica	Eyebrow	Expression
Occipitalis	Occipital bone	Galea aponeurotica	Moves scalp forward and backward

Buccinator	Alveolar processes of mandible and maxillary bone	Orbicularis oris and skin at angle of mouth	Mastication
Orbicularis oculi	Encircles eye	Encircles orbit and extends within eyelid	Closes eye
Orbicularis oris	Encircles mouth	Skin surrounding mouth and at angle of mouth	Closes mouth
Masseter	Zygomatic arch	Mandible	Mastication
Temporalis	Temporal fossa	Coronoid process and ramus of mandible	Mastication
Zygomaticus	Zygomatic bone	Corner of mouth	Smiling
Neck			
Sternocleidomastoid	Manubrium of sternum and median portion of clavicle	Mastoid process of temporal bone	Rotation and flexion of neck vertebrae
Back			
Latissimus dorsi	Vertebral column	Humerus	Extends at shoulder joint
Trapezius	Occipital bone and vertebral column (7 cervical vertebrae and all thoracic vertebrae)	Clavicle and scapula	Rotates, elevates, depresses, adducts, and stabilizes scapula
Pectoral girdle			
Pectoralis major	Sternum, clavicle, upper 6 ribs, and aponeurosis of external oblique muscle	Medial margin of tubercular groove of humerus	Flexion, rotation, and adduction of shoulder joint
Shoulder			
Deltoid	Lateral third of clavicle, acromion process, and spine of scapula	Deltoid tuberosity of humerus	Abduction, flexion, rotation, and extension of arm, using coordinated fibers: entire muscle (abduction); interior fibers (flexion); medial fibers (rotation); posterior fibers (extension); and lateral fibers (rotation) of arm
Abdominal wall			
External abdominal	External surface of lower 8 ribs	Aponeurosis to linea alba and anterior	Stabilizes, protects, and supports internal viscera, compresses abdominal cavity, assists in flexing

oblique		half of iliac crest	and rotating vertebral column
Internal abdominal oblique	Inguinal ligament, iliac crest, and lumbodorsal fasciae	Linea alba, pubic crest, and last 3 ribs	Stabilizes, protects, and supports internal viscera, compresses abdominal cavity, assists in flexing and rotating vertebral column
Transversus abdominis	Inguinal ligament, iliac crest, lumbodorsal fasciae, and costal cartilage of last 6 ribs	Linea alba and pubic crest	Stabilizes, protects, and supports internal viscera, compresses abdominal cavity, assists in flexing and rotating vertebral column
Rectus abdominis	Pubic crest and symphysis pubis	Xiphoid process of sternum and costal cartilage of ribs 5–7	Stabilizes, protects, and supports internal viscera
Thorax			
Diaphragm	Xiphoid process of sternum, inner surface of lower 6 ribs, and lumbar vertebrae	Central tendon of diaphragm	Pulls central tendon downward, increasing size of thoracic cavity and causing inspiration; separates thoracic and abdominal cavities
External intercostals	Inferior border of rib above and costal cartilage	Superior border of rib below rib of origin	Elevates ribs, aiding inspiration
Internal intercostals	Superior border of rib below and costal cartilage	Inferior border of rib above rib of origin	Depresses ribs, aiding inspiration
Arm			
Biceps brachii	Long head: Tubercle above glenoid fossa, Short head: Coracoid process of scapula	Tuberosity of radius	Flexion at elbow joint
Triceps brachii	Long head: Infraglenoid tubercle of scapula, Lateral head: Posterior surface of humerus above radial groove, Medial head: Posterior surface of humerus above radial groove	Olecranon process of ulna	Extension of elbow joint
Flexor carpi radialis	Medial epicondyle of humerus	2nd to 3rd metacarpals	Flexor of wrist, abducts hand
Flexor carpi ulnaris	Medial epicondyle of humerus and olecranon process	Pisiform bone and base of 5th metacarpal	Flexor of wrist, adducts hand
Supinator	Lateral epicondyle of humerus and proximal end of ulna	Proximal end of radius	Supinates forearm
Extensor carpi ulnaris	Lateral epicondyle of humerus	Base of 5th metacarpal	Extends and adducts wrist

Extensor carpi radialis longus	Lateral supracondylar ridge of humerus	Base of 2nd metacarpal	Extends and abducts wrist
Extensor carpi radialis brevis	Lateral epicondyle of humerus	Dorsal surface of 3rd metacarpal	Extends and abducts wrist, steadies wrist during finger flexion
Thigh, Anterior (Front)			
Rectus femoris (part of quadriceps)	Acetabulum of coxal bone	Tibial tuberosity via patella	Extends knee and flexes thigh at hip
Vastus lateralis (part of quadriceps), vastus medialis, vastus intermedialis (part of quadriceps)	Femur	Tibia	Extends knee
Sartorius	Anterior superior iliac spine	Proximal medial surface of tibia	Flexes at knee and laterally rotates thigh
Adductors	Pubis	Femur	Adduction and flexion at hip joint and lateral rotation of thigh
Gracilis	Pubis, symphysis pubis, and pubic arch	Medial surface of tibia	Adduction of thigh and rotation of leg
Thigh, Posterior (Back)			
Biceps femoris (part of hamstrings)	Long head: Ischial tuberosity, Short head: Linea aspera	Fibula and lateral condyle of tibia	Flexion at knee, extension of thigh
Semimembranosus (part of hamstrings)	Ischial tuberosity	Medial condyle of tibia	Flexion at knee, extension of thigh
Semitendinosus (part of hamstrings)	Ischial tuberosity	Proximal end of tibia	Flexion at knee, extension of thigh
Leg, Posterior			
Gastrocnemius	Medial and lateral condyles of femur	Calcaneus by Achilles (calcaneal) tendon	Flexion at knee and plantar flexion
Soleus	Posterior third of fibula and middle third of tibia	Calcaneus by Achilles tendon	Plantar flexion
Hip			
Gluteus maximus	Dorsal ilium, sacrum, and coccyx	Gluteal tuberosity of	Extends thigh and laterally rotates thigh

42 A muscle's name can be based on four things. Which one of those things was used to name these three muscles: the latissimus dorsi, the rectus abdominis, and the serratus anterior?

- a. Shape
- b. Attachment
- c. Figurative resemblance
- d. Function

43 The origin of the biceps brachii would best include which of the following?

- a. Scapula
- b. Clavicle
- c. Fibula
- d. Ulna

44-48 Match the origins and insertions for the following muscles.

- a. The pubis and the femur
- b. The femur and the calcaneus
- c. The ilium and the tibia
- d. The ischium and the tibia
- e. The pubis and the tibia

44. _____ Semimembranosus

45. _____ Gracilis

46. _____ Sartorius

47. _____ Gastrocnemius

48. _____ Adductors

49-53 Match the muscles with their actions.

- a. Rotation of scapula
- b. Flexion of leg at knee joint
- c. Extension at shoulder joint
- d. Mastication
- e. Flexion of forearm

49. _____ Semitendinosus

50. _____ Temporalis

51. _____ Biceps brachii

52. _____ Latissimus dorsi

53. _____ Trapezius

54-58 Match the muscles with their locations.

- a. Head
- b. Abdomen
- c. Back
- d. Neck
- e. Thigh

54. _____ Latissimus dorsi

55. _____ Internal oblique

56. _____ Quadriceps

57. _____ Masseter

58. _____ Sternocleidomastoid

59 What “two-headed” muscle is found in both the arm and the leg?

- a. Quadriceps
- b. Triceps

- c. Biceps
- d. Abdominals
- e. Gluteus

60 This muscle divides the thoracic cavity from the abdominal cavity.

- a. Diaphragm
- b. External oblique
- c. Transversus abdominis
- d. Rectus abdominus
- e. Internal oblique

61 Which of the following is *not* one of the muscles referred to as hamstrings?

- a. Biceps femoris
- b. Gracilis
- c. Semimembranosus
- d. Semitendinosus

Answers to Questions on Muscles

The following are answers to the practice questions presented in this chapter.

- ① Which of the following is *not* a true statement? **c. Sphincters hold fibers together to form muscles.** Sphincters are rings of muscle fibers (cells) used as a control mechanism.
- ② What happens during myogenesis and morphogenesis? **d. Embryonic muscles form during myogenesis, and muscles form into internal organs during morphogenesis.**
- ③ Why do your muscles shiver when you're cold? **b. To generate heat.** Chemical reactions in muscles result in heat, which is also why you stamp your feet and jog in place when you're cold.

- ④ Name the cellular unit in muscle tissue. **c. Fiber.** When it comes to muscle, fibers and cells are the same thing.
- ⑤ What defines a state of tension present to a degree at all times? **b. Tonus.** That's the elusive muscle "tone" for the flabby amongst us.
- ⑥ It's possible to completely relax every muscle in the body. **b. False.** If every muscle in the body were to relax, the heart would stop beating and food would stop moving through the digestive system.
- ⑦ Exercise forms new muscle fibers. **b. False.** Exercise can't form new fibers, it stimulates protein synthesis within the cells, causing them to enlarge.
- ⑧ What type of muscle tissue lacks cross-striations? **b. Smooth.** Fascia is not muscle tissue and contracting muscle tissue is a made-up term.
- ⑨ What does a sarcolemma do? **a. It receives and conducts stimuli for skeletal muscle tissue.** Other answers relate to other types of muscles, so if you associate "skeletal" with "sarcolemma," you'll stay on the right track.
- ⑩ Which muscle type appears only in a single organ? **c. Cardiac.** And that sole organ is the heart.
- ⑪ Intercalated discs **b. play a role in moving electrical impulses through the heart.** There's evidence that these structures help keep the heart synchronized.
- ⑫ **b. Smooth muscle**
- ⑬ **c. Cardiac muscle**
- ⑭ **d. Skeletal muscle**
- ⑮ Which of these terms doesn't belong in the following list? **d. Sarcolemma.** This is the cell membrane encasing the myofibrils. All the other answer options refer to various protein structures.
- ⑯ During contraction, what does *not* occur? **a. Actin and myosin filaments shorten.** Quite the opposite: Muscle fibers shorten, actin and myosin

merely overlap.

- 17) A weak stimulus causes a muscle fiber to contract only partway. **b. False.** Muscle contractions are all-or-nothing; there's no such thing as a partial contraction.
- 18) Which of the following steps of muscle contraction requires the breakdown of ATP? **e. myosin heads cocking back to their original position.** While ATP must be present to cause the myosin heads to let go, the energy it contains is not used until the heads cock back.
- 19) **a. Titin**
- 20) **e. H zone**
- 21) **c. I band**
- 22) **d. A band**
- 23) **b. Z line**
- 24) Which statement best characterizes a large motor unit? **e. One neuron stimulate many fibers for a powerful movement.**
- 25) The steps should go something like this:
- Neuron releases ACh**
 - ACh spreads across the synaptic cleft**
 - ACh binds to the motor end plate**
 - Muscle fiber generates impulse**
 - Impulse spread through sarcolemma then through the t-tubules**
 - Impulse stimulates the SR which releases calcium ions**
 - Calcium ions bind to troponin**
 - Tropomyosin (or t-t complex) slides around, uncovering the binding sites**
- 26) Areolar tissue that surrounds fibers: **e. Endomysium.**

- 27) Bundles of muscle fibers: **d. Fascicles.**
- 28) Connective tissue that wraps muscle fibers into bundles: **a. Perimysium.**
- 29) Connective tissue that creates a functional muscle **c. Epimysium.**
- 30) Flat, sheet-like tendon that serves as insertion for a large, flat muscle: **b. Aponeurosis.**
- 31) Which of the following muscle tissues are striated? **d. I and II:** While all three tissues have visible, length-wise stripes in the fibers, the striations are the stripes that run the length.
- 32) While structurally different, all three muscle fiber types contract their myofibrils utilizing the sliding-filament model. **a. True:** The difference is how the myofibrils are arranged within a cell.
- 33) What do you call muscles that tend to counteract or slow an action? **a. antagonists**



TIP

The muscles are against the action, so think of them as antagonistic.

- 34) What creates a strong muscle contraction? **d. High frequency stimulatory nerve impulses.** The first two answers relate to types of muscle contractions, not strength.
- 35) As stimulation increases, more motor units are triggered to contract: **c. Recruitment.**
- 36) When twitches occur before the previous one ends, generating more force: **a. Summation.**
- 37) Stimulation occurs so frequently that no relaxation occurs at all: **d. Tetanus.**
- 38) Constant, sustained contraction of some fibers within a muscle: **b. Tone.**

- 39) Which of the following in [Figure 8-6](#) is a Class II lever? **b. Pectoralis major.** With Class II, remember to look for wheelbarrow-like leverage.
- 40) Which muscles would provide the force in a Class III lever? **a. Biceps brachii.** It's the Popeye weight-lifting class, after all.
- 41) Which of the following would produce a wide range of movement with speed while sacrificing power? **b. Long weight arm, short power arm.** The longer the weight arm, the greater the range of action and speed *but* the less power there is.
- 42) A muscle's name can be based on four things. Which one of those things was used to name these three muscles: the latissimus dorsi, the rectus abdominis, and the serratus anterior? **a. Shape.** It's a matter of memorizing a bit of Latin: *latissimus* stems from the Latin word for "wide," *rectus* from the Latin word for "straight," and *serratus* from the Latin word for "notched" or "scalloped."
- 43) In humans, the origin of the biceps brachii would best include which of the following? **a. Scapula.** After you know that your biceps brachii are in your upper arm, you just have to remember the name of the origin bone. The Latin meaning of *scapulae* is, literally, "shoulder blades," so perhaps a better memory tool would be that the scapula looks vaguely like a spatula.
- 44) Semimembranosus: **d. The ischium and the tibia.**
- 45) Gracilis: **e. The pubis and the tibia.**
- 46) Sartorius: **c. The ilium and the tibia.**
- 47) Gastrocnemius: **b. The femur and the calcaneus.**
- 48) Adductors: **a. The pubis and the femur.**
- 49) Semitendinosus: **b. Flexion of leg at knee joint.**
- 50) Temporalis: **d. Mastication**
- 51) Biceps brachii: **e. Flexion of forearm.**

- 52) Latissimus dorsi: **c. Extension at shoulder joint.**
- 53) Trapezius: **a. Rotation of scapula.**
- 54) Latissimus dorsi: **c. Back.**
- 55) Internal oblique: **b. Abdomen.**
- 56) Quadriceps: **e. Thigh.**
- 57) Masseter: **a. Head.**
- 58) Sternocleidomastoid: **d. Neck.**
- 59) What “two-headed” muscle is found in both the arm and the leg? **c. Biceps.** Your biggest clue: the “bi” part of biceps. Although people usually think of the biceps brachii in the arm, you can’t forget about the biceps femoris at the back of the thigh.
- 60) This muscle divides the thoracic cavity from the abdominal cavity. **a. Diaphragm.** And without it, you couldn’t breathe.
- 61) Which of the following is not one of the muscles referred to as hamstrings? **b. Gracilis.** The other three answer options all are listed as hamstring muscles.

Part 3

Mission Control: All Systems Go